UNIVERSITY OF GONDAR

COLLEGE OF INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

WEB-BASED COMPUTER LABORATORY MANAGEMENT SYSTEM FOR UOG

INDUSTRIAL PROJECT

PROJECT PARTICIPANTS	ID NUMBER
1. ASMAMAW DEGU	GUR/02284/15
2. GETANEH SITAYEHU	GUR/01812/14
3. ASCHALEW TEGEGNE	GUR/02263/15
4. TESFA ZERFU	GUR/02271/15
5. YALEMWORK AYENEW	GUR/ 00769/14
6. SAMSON TEKLAY	GUR/02582/15

Advisor: Mr. Amare Tesfa

Gondar, Ethiopia

Date: March 2026

APPROVAL SHEET

This Group Project entitled Web-based Computer Laboratory Management System for UoG has been read and approved as meeting the preliminary project requirements of the Department of Information Technology in partial fulfillment of the requirements for the award of a Bachelor of Science degree in Information Technology, UoG, Gondar, Ethiopia.

Approved by:

No.	Name	Role	Signature	Date
1	Mr. Amare Tesfa	Project Advisor	_____	_____
2	_____	Examiner	_____	_____
3.		Examiner		
4.		Examiner		

Table of Contents

APPROVAL SHEET i
LIST OF TABLES iv
LIST OF FIGURES v
LIST OF ACRONYMS vi
ACKNOWLEDGEMENTS vii
EXECUTIVE PROJECT SUMMARY viii
CHAPTER ONE: INTRODUCTION 1
1.1 Background of the Project 2
1.2 Statement of the Problem and Justification 3
1.3 Objective of the Project 3
1.3.1 General Objective 4
1.3.2 Specific Objectives 4
1.4 Scope of the Project 4
1.5 Limitations of the Project 5
1.6 System Development Methodology 5
1.6.1 System Development Paradigm 6
1.6.2 System Development Tools 6
1.7 Significance of the Project 7
1.8 Beneficiaries of the Project 8
1.9 Feasibility Study 8
1.9.1 Technical Feasibility 8
1.9.2 Operational Feasibility 9
1.9.3 Economic Feasibility 9
1.9.4 Legal Feasibility 9
1.10 Project Schedule 10
1.11 Budget Plan 11
CHAPTER TWO: REQUIREMENT ANALYSIS 12
2.1 CURRENT SYSTEM DESCRIPTION 12
2.1.1 Major Functions of the Current System 12
2.1.2 Problem of Existing System 14
2.2 Requirement Gathering 14
2.2.1 Requirement Gathering Methods 14
2.2.2 Business Rules 15
2.3 Proposed System Description 16

2.3.1 Overview 17
2.3.2 Functional Requirements 19
2.3.3 Non- Functional Requirements 21
2.3.3.1 Performance 21
2.3.3.2 Scalability 21
2.3.3.3 Availability 21
2.3.3.4 Reliability 22
2.3.3.5 Maintainability 22
2.3.3.6 Security 22
2.3.3.7 Environmental 22
2.3.3.8 Usability 23
2.3.3.9 Interoperability 23
CHAPTER THREE: SYSTEM MODEL 24
3.1 Scenario 24
3.1.1 Use Case Model 25
3.1.2 Use Case Diagram 26
3.1.3 Use Case Descriptions 28
3.1.4 Activity Diagram 43
3.1.5 Object Diagram 45
3.1.6 Data Dictionary 47
3.1.7 Class Model 49
3.1.7.1 Class Diagram 49
3.1.8 Dynamic Modeling 51
3.1.8.1 Sequence Diagram 51
3.1.8.2 State Chart Diagram 54
3.1.9 User Interface Prototype 56
CHAPTER FOUR: SYSTEM DESIGN 59
4.1 Introduction 59
4.2 Current Software Architecture 59
4.3 Proposed Software Architecture 62
4.3.1 System Decomposition 63
4.3.2 Component Diagram 65
4.3.3 Hardware/Software Mapping 66
4.3.4 Persistent Data Management 67
4.3.5 Access Control and Security 68
4.3.6 Package Diagram 70
4.3.7 Deployment Diagram 71

CHAPTER FIVE: CONCLUSION AND RECOMMENDATION 73
CONCLUSION 73
RECOMMENDATION 74
REFERENCES 75
Appendix: Group Discussion Methods Requirements Validation 76

LIST OF TABLES

Table 1.1: System Development Tools 7
Table 1.2: time frame 10
Table 1.3: budget plan 11
Table 2.1: Summary of functional requirements 21
Table 3: Actor Specification 26
Table 3.1: Description of Login 28
Table 3.2: Description of Book Workstation 29
Table 3.3: Description of Reserve Lab 30
Table 3.4: Description of Report Fault 31
Table 3.5: Description of Manage Inventory 32
Table 3.6: Description of Conflict Detection 33
Table 3. 7: Description of Manage User Account 34
Table 3.8: Description of the Generate Report 35
Table 3.9: Description of Monitor Attendance 36
Table 3.10: Description of Download Material 37
Table 3.11: Description of Logout 38
Table 3.12: Description of Registration 39
Table 3.13: Description of check equipment status 40
Table 3.14: Description of View Lab Availability 41
Table 3.15: Description of Upload Material 42
Table 3.16: Data Dictionary 49
Table 4.1: Access Control and Security 70
Table 4.2: Verification Sample 76

LIST OF FIGURES

Figure 2: CLMS for the UOG Hierarchical Model 13
Figure 2.1: overview of the proposed system diagram 18
Figure 3.1: Use Case Diagram 27
Figure 3.2: Activity Diagram of User Management 43
Figure 3.3: Activity Diagram of Authentication 43
Figure 3.4: Activity Diagram of Maintenance 44
Figure 3. 5: Activity Diagram of Inventory Tracking 44
Figure 3. 6: Activity Diagram of Lab Booking 45
Figure 3.7: Activity Diagram of Reporting 45
Figure 3.8: Object Diagram 46
Figure 3.9: Class Diagram 50
Figure 3.10: Login Sequence diagram 52
Figure 3.11: Report Sequence diagram 53
Figure 3.12: Reserve Sequence diagram 53
Figure 3.13: Maintenance Sequence diagram 54
Figure 3.14: Booking Sequence diagram 54
Figure 3.15: State Chart Diagram of Student 55
Figure 3.16: State Chart Diagram of Teacher 55
Figure 3.17: State Chart Diagram of Admin 56
Figure 3.18: State Chart Diagram of a Lab Technician 56
Figure 3.19: Login/Register User Interface 57
Figure 3.20: Admin Dashboard User Interface 58
Figure 4.1: Current Software Architecture 62
Figure 4.2: System Architecture Model for the Proposed System 62
Figure 4.3: Decomposition Diagram 64
Figure 4.4: Component Diagram 65
Figure 4.5: Hardware/Software Mapping 67
Figure 4.6: Database Schema 68
Figure 4.7: Package Diagram 71
Figure 4.8: Deployment Diagram 72

LIST OF ACRONYMS

BR – Business Rule

CLMS – Computer Laboratory Management System

CSS – Cascading Style Sheets

DBMS – Database Management System

GC – Health Science College (GC Campus)

HTML – Hypertext Markup Language

ICT – Information and Communication Technology

ID – Identifier

INT – Integer

JS – JavaScript

MTTF – Mean Time To Failure

MTTR – Mean Time To Recovery

OOP – object-oriented paradigm

PC – Personal Computer

PHP – Hypertext Preprocessor

RAD – Rapid Application Development

RBAC – Role-Based Access Control

RDBMS – Relational Database Management System

SPMP – Software Project Management Plan

SQL – Structured Query Language

SRS – Software Requirement Specification

UC – Use Case

UML – Unified Modeling Language

UoG – University of Gondar

VARCHAR – Variable Character

VS Code – Visual Studio Code

XSS – Cross-Site Scripting

ACKNOWLEDGEMENTS

First and foremost, we would like to thank Almighty God for giving us the strength, wisdom, and perseverance to successfully complete this project.

We would like to express our sincere gratitude to our advisor, Mr. Amare Tesfa, for his invaluable guidance, continuous support, and encouragement throughout this project.

We also extend our thanks to College of Informatics, University of Gondar, for providing the necessary resources and a conducive learning environment. Our appreciation goes to the staff and lab coordinators of Tewodros Campus for their cooperation during the requirement-gathering process.

EXECUTIVE PROJECT SUMMARY

This project proposes the development of a Web-Based Computer Laboratory Management System (CLMS) for the University of Gondar (UoG). Nowadays, many application systems use an online system to interact with users because online systems provide many advantages over manual systems. The Computer Laboratory Management System is an example of an online system that will help lab administrators manage laboratories more efficiently. The system is designed to address the limitations of the existing manual, paper-based method currently used to manage computer laboratories across the university's four campuses: Maraki, Atse Tewodros, Atse Fasil, and Health Science College (GC). The current manual approach is slow, prone to errors, and unable to cope with the increasing demand.

The proposed CLMS is a secure, centralized, and user-friendly online platform that manages various laboratory activities such as student and instructor registration, laboratory scheduling, workstation booking, resource assignment, inventory tracking, and maintenance documentation. The system incorporates role-based security to ensure that only authorized individuals including students, instructors, lab assistants, campus administrators, and super administrators can access functions appropriate to their roles. Key features include real-time tracking of laboratory resources, automated report generation, online booking with conflict detection, digital fault reporting with maintenance ticketing, automated notifications, equipment tracking and maintenance logging, resource monitoring, and secure access control.

By replacing the manual record-keeping process, the system will reduce scheduling conflicts, minimize equipment downtime, improve accountability, and increase user satisfaction across all campuses. Developed using PHP, MySQL, HTML5, CSS3, and JavaScript with a three-tier architecture, the system will introduce modern laboratory management practices and support the university’s vision of becoming one of the top research universities in Africa by 2030.

CHAPTER ONE: INTRODUCTION

In the current educational landscape, computer laboratories play a significant role and are transforming from technology rooms to hubs or centers. An effective way to manage resources is essential to maximize their benefits. However, at the University of Gondar (UoG), laboratory management is currently conducted through paper-based methods and physical visits. The management of computer laboratories at UoG is primarily done through manual logbooks and verbal discussions. This practice creates duplicate operations and leads to extended equipment maintenance times. Furthermore, the current system does not provide efficient monitoring capabilities or real-time analysis for laboratory managers. This results in wasted financial resources, loss of valuable student learning time, and underutilization of laboratory assets.

These challenges directly affect the university's three core mission pillars: teaching and learning, research, and community service. Inefficient laboratory management limits students' access to practical learning environments, delays academic activities, and reduces the effectiveness of technology-supported instruction. It also restricts research activities that depend on reliable and well-maintained laboratory resources. Moreover, poor resource coordination weakens the university's capacity to provide quality community service programs that rely on laboratory infrastructure and training services.

In response to these operational challenges and to bridge the gap between the university's technological vision and its current operations, this project proposes the development of a Computer Laboratory Management System (CLMS). The proposed CLMS will include user management features for registration and login of students, instructors, and administrators. It will provide lab scheduling and booking capabilities, enabling real-time reservation of computers and laboratory sessions while preventing conflicts. Equipment tracking and maintenance functions will allow automated logging of hardware and software status and maintenance schedules. The system will also include resource monitoring and reporting to enable real-time tracking of laboratory usage and the generation of analytical reports for managers. Notifications and alerts will remind users of maintenance schedules, reservations, or overdue usage, while security and access control mechanisms will enforce role-based access to ensure authorized usage of laboratory resources. The CLMS will operate on a centralized, database-driven system to ensure reliable management of all UoG laboratory resources, reducing duplication and manual errors. By implementing these features, the CLMS will enhance the efficiency of laboratory management, optimize the use of UoG laboratory resources, and support the university's core missions of teaching and learning, research, and community service.

1.1 Background of the Project

The University of Gondar was founded in 1954. It is a non-profit public higher education institution located in the urban setting of the larger city of Gondar. The institution has branches in the following locations: Maraki, Atse Tewodros, Atse Fasil, and Health Science College (GC).

Officially accredited by the Ministry of Education of Ethiopia, the University of Gondar (UoG) is a large educational institution that educates over 15,000 students in various fields of study at the undergraduate, graduate, and doctoral levels across all its campuses, producing qualified citizens. The university has a vision: "University of Gondar aspires to be one of the top ten research universities in Africa by 2030," and a mission: "University of Gondar provides quality education, research, and community engagement responsive to national and global needs through entrepreneurial culture." In order to fulfill this vision and mission, it is committed to offering courses and programs leading to officially recognized higher education degrees in various fields of study.

The university maintains more than 33 computer laboratories with over 1,670 desktop computers for various purposes, including instruction, project work, seminars, and research across all its campuses. However, the management of these resources is entirely manual, requiring students and staff to be physically present to access services. This manual approach creates significant difficulties for the institution in terms of resource sharing, financial efficiency, time management, human resource allocation, meeting student needs, reporting, and prompt problem resolution. In an era where administration and management have become digital for efficiency, a Computer Laboratory Management System (CLMS) offers an effective solution.

The Computer Laboratory Management System (CLMS) is a web-based system proposed for implementation in the University of Gondar computer laboratories. Due to manual tracking, minor technical problems often go unreported for days, and equipment downtime can extend to weeks. In an environment where the student-to-computer ratio is already high, even 10% of workstations being non-functional directly affects learning quality for hundreds of students. Furthermore, the absence of a digital audit trail creates accountability gaps that result in asset misplacement and underutilization of the university's ICT investment. This disparity between the university's technological vision and the current administrative reality necessitates the development of a web-based CLMS. The project will replace manual, inefficient processes with a streamlined digital platform, enabling the University of Gondar to fulfill its mission of creating a modern, equitable, and resource-efficient learning environment.

1.2 Statement of the Problem and Justification

UoG works towards excellence in technology, but the paper-based practice in managing its computer facilities creates a big gap in its operations. This was particularly true of the lack of any digital blueprint concerning laboratory timetabling, equipment inventory, or maintenance tracking. Without the automation of such a system, schedules orchestrated through physical logbooks lead to frequent double-bookings and session overlaps, events that are coupled with overcrowding in laboratories and the cancellation of major practical classes.

The problem is severe and extends to levels of inventory control and maintenance protocols: as of now, no current mechanism exists to monitor the functionality of individual workstations in real-time; technical breakdowns are only identified reactively, and many times, requests for repairs get lost within administrative papers. This also makes functional computers idle for weeks; as a result, there is a degradation of costly ICT equipment, as there is no accountability regarding the condition of such equipment.

The manual system makes proper resource sharing and reporting difficult. These reporting inefficiencies lead to substantial financial waste and lost learning time. The gap between what is required to achieve the university's mission and what actually occurs on a daily basis will continue to widen without a comprehensive Computer Laboratory Management System.

Managing tasks across multiple locations is difficult under the manual system**:**

1. It is difficult to manage laboratory tasks across multiple locations using a manual system. For example, users such as staff and students must go to the resource room to book a laboratory.

2. Recorded data can easily be lost when lab administrators use the manual system. This issue may create additional problems for lab administrators when retrieving important records.

3. Lab administrators find it difficult to troubleshoot and repair laboratory equipment when devices are damaged because the manual system does not provide an efficient way to track maintenance activities.

Therefore, implementing a Computer Laboratory Management System (CLMS) is essential to ensure efficient scheduling, proper maintenance, accurate inventory tracking, and real-time monitoring of laboratory resources. Such a system will minimize operational inefficiencies, reduce financial waste, maximize the utilization of laboratory equipment, and enhance the quality of teaching, learning, and research at the University of Gondar.

1.3 Objective of the Project

1.3.1 General Objective

To develop a web-based Computer Laboratory Management System for the University of Gondar.

1.3.2 Specific Objectives

1. To identify and gather the requirements of the CLMS.
2. To design the CLMS for better use.
3. To develop an interactive and user-friendly interface for the CLMS.
4. To develop the functional modules of the CLMS.
5. To integrate all modules into a centralized and secure web-based platform.
6. To evaluate the performance of the CLMS.

1.4 Scope of the Project

The Functional Scope of this system is limited to six modules only, which are User Management, Lab Scheduling & Booking, Inventory & Asset Management, Maintenance & Fault Management, Reporting & Analytics, and Notification & Alerts. These modules define the core functionalities that the system will provide to manage laboratory operations effectively.

The User Scope is established by the collaborative and functional relationships in the laboratory environment. The system mediates the workflow between various stakeholders for seamless academic delivery and resource accounting. For example, data integration, Resource sharing, easily fixing problems, distinguishing lab safety, and the system supports the student-teacher association by enabling students to access the resources and schedules controlled by the teacher for the particular practical classes. Again, the system fills the gap between the teachers and the lab technicians, as the teacher relies on the technicians for the equipment availability and arrangement for the respective classes. Beyond this, the system coordinates the work of lab technicians and administrators, where it offers a virtual environment to report breakdowns in equipment and changes in inventories that fall within the administrative domain. With its role-based access control, CLMS ensures each user interaction falls squarely within specific academic and administrative roles.

Geographically, this first implementation covers the computer laboratories within the Tewodros, Maraki, Fasil, and GC campuses of the University of Gondar. Although the project will be implemented initially within the catchment area specified, its modular architecture accommodates scaling up and replication with ease for other university clusters, departments, and computer centers in subsequent times.

1.5 Limitations of the Project

The proposed system aims to address major challenges in computer laboratory management; certain aspects fall outside the scope of this project. The system will not automate activities that require physical human intervention, such as hardware repairs, physical access control, procurement processes, or network-level security enforcement. It will not manage university-wide ICT assets outside laboratory spaces, nor will it integrate with external enterprise systems such as university-wide learning management systems, attendance systems, or biometric authentication during this development phase. The proposed system uses only the English language.

1.6 System Development Methodology

System development methodologies provide structured frameworks for planning, designing, implementing, and maintaining software systems. Common methodologies include the Waterfall Model, Agile, Rapid Application Development (RAD), the Spiral Model, and the V-Model. The Waterfall and V-Models are linear and sequential, emphasizing thorough documentation, but are inflexible to requirement changes. RAD focuses on rapid prototyping and user feedback but may compromise detailed planning, while the Spiral Model combines iterative development with risk analysis, suitable for large or high-risk projects but often complex and costly. In contrast, agile methodology is adaptive and iterative, promoting incremental delivery, continuous feedback, and stakeholder collaboration, making it ideal for projects with evolving requirements.

For the development of the Computer Laboratory Management System (CLMS) at the University of Gondar, Agile was selected as the most appropriate methodology. It provides a structured framework while incorporating iterative development principles, allowing the system to be developed incrementally in small, testable modules, such as User Management, Lab Scheduling & Booking, Inventory & Asset Management, Maintenance & Fault Management, Reporting & Analytics, and Notification & Alerts. While Iterative Development enables incremental delivery and feedback, it is considered an approach, not a complete methodology, because it does not define project roles, processes, or management practices. Agile, however, provides these structures, making it suitable for systematic development and deployment of the CLMS.

1.6.1 System Development Paradigm

A system development paradigm is an approach or model used to design, develop, and implement software systems, providing a framework of principles, methods, and tools. Two common paradigms are the structural (procedural) paradigm and the object-oriented paradigm (OOP). The structural paradigm focuses on functions and processes, keeping data and procedures separate, which makes modeling real-world entities and maintaining large systems more difficult. In contrast, OOP focuses on objects that combine data and behavior, supports encapsulation, inheritance, and reusability, and allows easy modeling of real-world entities.

For the Computer Laboratory Management System (CLMS) at the University of Gondar, OOP is selected because it enables clear representation of students, instructors, lab assistants, administrators, laboratory equipment, and schedules as objects. It improves modularity, maintainability, security, and scalability, making the system easier to manage, extend, and maintain in the future.

1.6.2 System Development Tools

The technology used in building the project will be state-of-the-art and open source, reflecting which technology the developers feel comfortable with and what the institution can afford. On the backend technology, PHP will be used for building the application logic, authentication, and communication with the database. MySQL will be the relational database management system, which holds users, schedules, and hardware inventory.

In terms of development, we will utilize the help of XAMPP so as to create a ready-made setup solution including Apache, MySQL, and PHP. But should we require server-side programming through JavaScript code to handle such responsibilities?

Finally, with regard to the front-end section, through the help of HTML5, CSS3, and JavaScript libraries, we would be developing the user interface so as to provide a contemporary and user-friendly look.

Tool Type	Selected Tools	Purpose / Justification
Programming Languages	HTML5, CSS3, JavaScript, PHP	- HTML/CSS: For webpage structure and styling. - JavaScript: For creating interactive, responsive user interfaces. - PHP: For back-end logic and database connectivity.
Database Management	MySQL	A reliable, open-source database system that integrates seamlessly with PHP to organize and manage system data.
Development Environment	Visual Studio Code, XAMPP	- VS Code: A powerful code editor for writing and debugging. - XAMPP: Provides a local server environment (Apache, MySQL, and PHP) for development and testing.
Documentation Tools	Microsoft Word	Used for writing technical reports, user manuals, and all project documentation.

Table 1.1: System Development Tools

1.7 Significance of the Project

The implementation of the Computer Laboratory Management System (CLMS) is a vital step toward modernizing the University of Gondar's academic infrastructure. Currently, the university faces significant pain points due to manual management, such as data redundancy, loss of Resource sharing, quick fix problem, a lack of distinction between lab safety, high error rates in paper logbooks, and a heavy administrative burden. The CLMS solves these issues by automating the entire process. It removes the need for multiple data entries and eliminates the risk of human error associated with manual record-keeping, ensuring that laboratory data is centralized, accurate, and easily retrievable.

The system provides specific, high-value benefits to all key stakeholders. For students, it guarantees fair and organized access to computers for assignments and digital research. Lecturers gain a structured platform to plan and schedule practical classes with real-time visibility of lab availability. Laboratory assistants benefit most significantly, as the system frees them from tedious manual entries, allowing them to focus on technical assistance and maintenance. For administrators, the system provides a consolidated dashboard of ICT resources, enabling efficient resource sharing between different labs and campuses to prevent shortages in one area while

equipment sits idle in another. This data-driven oversight replaces "gut-feelings" with clear insights for better resource allocation and informed decision-making.

From an economic perspective, the project ensures the long-term financial sustainability of the university's investments. By tracking inventory and logging maintenance in real-time, the system prevents waste caused by lost equipment, machine downtime, and neglected repairs. This proactive approach maximizes the useful life of hardware assets and optimizes resource allocation. Ultimately, the CLMS reduces operational costs and ensures that the university extracts maximum value from its ICT infrastructure, positioning the University of Gondar as a leader in efficient academic management.

1.8 Beneficiaries of the Project

The project serves the whole academic and operational community at UoG. Students enjoy uninterrupted, easy access to lab computers for their assignments, online learning, and research. Instructors have a smooth and predictable process for scheduling and reservations, reducing delays and clashes in classes. Lab technicians have clearer workflows as they attend to user requests, log faults, and prioritize technical fixes. Management develops strategic insight into the manner of utilization and condition of equipment, and therefore, it allows them to turn guesses into proactive planning. University of Gondar will benefit from a better return on ICT investment, stronger accountability in resource management, and greater credibility in technology-enhanced education. In the end, the initiative supports the university’s goal of providing modern, equitable learning environments.

1.9 Feasibility Study

A feasibility study is an assessment done before starting a project to determine whether the project is practical, possible, and worth doing, considering technical, economic, legal, operational, and scheduling factors.

1.9.1 Technical Feasibility

The proposed system is quite feasible from a technical standpoint as it is seamlessly integrated within the current university infrastructure, whose technology platform is sound. As indicated, with PHP and MySQL, it is important to acknowledge that the technology platform is sound; therefore, it is more maintainable. One of the technical aspects of this proposed system is scalability, which is required to maintain increased data volume on many campuses. Moreover, it is sound from a technical standpoint as it is insensitive to those fluctuations of the internet, as it runs on the university’s local Intranet. The technical aspects of this proposed system, including sound authentication processes, data integrity managed by RDBMS, as well as browser-independent compatibility, entail that this proposed system is sound from a technical standpoint.

1.9.2 Operational Feasibility

The system is also highly operationally feasible since it serves the concrete daily administrative and academic needs of the University of Gondar. Rather than imposing additional complicated hurdles, the system actually places a digital structure on processes that users are performing manually, such as lab scheduling and fault reporting. This means a very low learning curve and high user adoption rates, whereby students, lecturers, and assistants can easily replicate their current tasks through a more efficient digital interface. A strong aspect that adds to the operational feasibility of the system involves resource sharing: administrators can gain a consolidated overview of equipment, thus enabling them to reallocate such idle resources between laboratories effectively. The system ensures laboratory operations are streamlined, transparent, and correctly integrated with the university's academic goals through the automation of conflict detection in scheduling and the provision of real-time maintenance tracking.

1.9.3 Economic Feasibility

From a cost-saving perspective, the proposed system is set to deliver significant cost-saving benefits compared to the current manual system in place. Organizing schedules and inventory through a manual system consumes a significant amount of labor. Additionally, poor management and a lack of visibility concerning equipment can lead to quick wear and tear and unnecessary purchases. By streamlining such functions and increasing the usage rate of available hardware, there is a significant reduction in labor hours and unnecessary purchases. Moreover, open-source development tools mean low costs. Development tools mean low costs.

1.9.4 Legal Feasibility

The Computer Laboratory Management System for UoG has been developed to strictly conform to Ethiopian legal standards and the University of Gondar's internal policies. In this case, full Intellectual Property rights over the custom-built system will be held by the University, and any open-source component parts that are integrated will be subject to licensing terms.

For the purpose of Data Privacy & Security, the system shall incorporate the best practices of encryption & access control in line with the Ethiopian Data Protection Proclamation & UoG IT guidelines. Additionally, the project also ensures Regulatory Compliance as per institutional norms in respect of accessibility, acquisition, & infrastructure. All obligations, timelines, & legal obligations of the project shall be aptly recorded in the form of formal agreements. Additionally, the title of the project is unique to the institution & does not have any trademark issues. Therefore, the project is feasible & completely in line with the institutional norms.

1.10 Project Schedule

we will discuss whether the Computer Laboratory Management System can be completed in accordance with the tight schedule laid down in the academic program. There are major risks in this schedule, which are mainly human and administrative and do not relate to technical aspects:

Time required by the development team to acquire the necessary skills;

A hard deadline constraint from the department, which can be restrictive in answering unexpected delays.

In order for the project to be completed on time, a real, feasible project schedule has been developed. The approximate time that will be taken for each of the project phases is presented in the table below. The project schedule has been developed in such a way that it fits into the academic calendar.

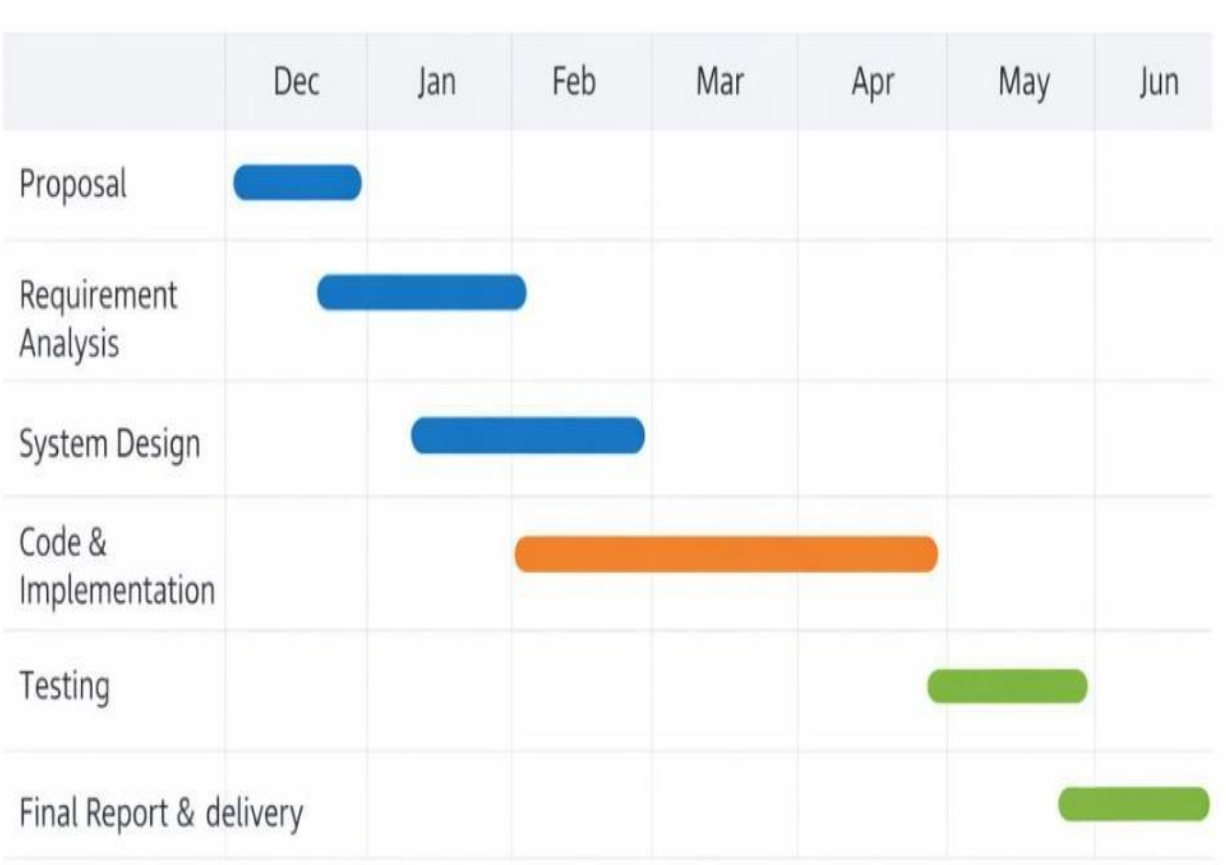

Table 1.2**:** time frame

1.11 Budget Plan

The project budget is planned with a focus on utilizing existing resources and open-source technologies to minimize costs. The proposed budget is set as a maximum cap and is designed to be sufficient for project completion. All costs have been evaluated for the project's entire lifecycle, and while expenditures may be reduced, the total budget cannot be increased in later stages. The detailed budget allocation is presented in the table below.

Project expenditures	Cost
1 Software	
Application software licenses	0
1.2 Operating system	0
1.3 Database user license	0
1.4 Additional license	0
1.6 local host server	0
2 Hardware	
2.1 PC's	0
2.5 Back-up device	500
3 Network cost	1000
3.1 Cabling or wireless LAN, or others	0
Others	500
Total	2000

Table 1.3**:** budget plan

CHAPTER TWO: REQUIREMENT ANALYSIS

2.1 CURRENT SYSTEM DESCRIPTION

2.1.1 Major Functions of the Current System

Currently, the computer laboratories' management system at UoG performs a variety of functions manually, which ideally should supervise activities related to the use of laboratories and the resources within them. First, the system records attendance and user access, requiring instructors or assistants to manually log each student's name, identification number, and entry time on cards attached to paper-based logbooks. Next, the scheduling of laboratories is handled by manually drafting and posting printed timetables on notice boards to communicate room allocations to students and staff. Inventory control, or hardware asset management across rooms, is conducted using physical ledgers to count system units and peripherals. Finally, work orders are raised by manually logging technical issues in a file or communicating them by word-of-mouth to the Lab Technical and Maintenance Unit. Overall, this manual system depends entirely on staff presence and the efficiency of paper records to keep the laboratories organized and functional.

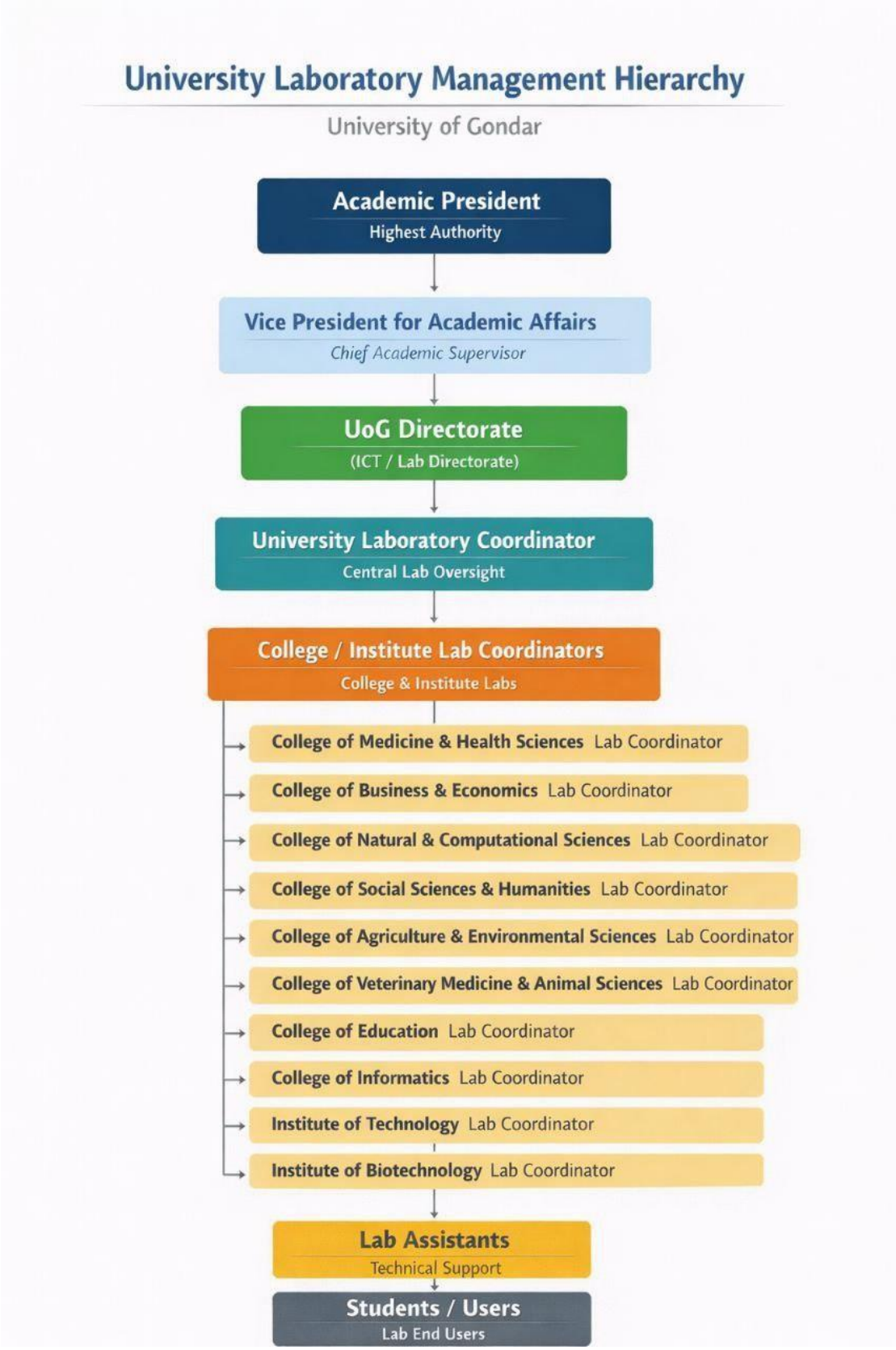

Figure 2**:** CLMS for the UOG Hierarchical Model

2.1.2 Problem of Existing System

The current management systems used at the University of Gondar's computer laboratories face several operational problems that affect efficiency. Reserving laboratory rooms is done manually using printed timetables on notice boards, which can lead to scheduling conflicts, double bookings, and miscommunication between students and lab coordinators. Attendance recording relies on physical logbooks, which are prone to damage, loss, or errors, making it difficult to maintain accurate records. Inventory and resource management are handled through physical files and ledgers, which are time-consuming and may cause delays in locating or transferring equipment between labs. Maintenance and fault reporting are done verbally or via written logs, often delaying repairs and increasing equipment downtime. Additionally, manual communication of schedule changes through notice boards or reports may not reach all users promptly, creating misunderstandings between lab coordinators and students. These problems collectively result in increased workload for staff, inefficient use of resources, and potential disruptions in laboratory operations, highlighting the need for a more automated and streamlined management system.

2.2 Requirement Gathering

Requirement gathering remains a core step in the development of the Computer Laboratory Management System, as it helps to identify the proposed system's requirements and align them with the needs of the University of Gondar. This process involves identifying the expectations of students, UOG technical staff, and laboratory coordinators in order to fill the gap between the university's daily activities and the technical development. By conducting requirement gathering, the project establishes a solid foundation for developing a user-friendly system aimed at improving data integrity and resource accountability within the university.

2.2.1 Requirement Gathering Methods

Developing an effective Computer Laboratory Management System for the University of Gondar requires an extensive approach to understanding user requirements and institutional workflows. To ensure the developed software is grounded in the university's reality, three methods of data collection were used:

Document Review: This involved a thorough examination of existing institutional records, laboratory manuals, and procedural guidelines. The focus was specifically on lab usage logbooks, student schedules, and equipment maintenance records. These documents provided a comprehensive understanding of the formal administrative demands and the official protocols governing the university's laboratories.

Observations: Direct observations were carried out in the university's computer laboratories during peak operational hours. These observations revealed that all current lab operations are conducted manually. It was noted that students must wait in long queues to register in paper logbooks, and instructors are required to visit technicians in person to manage laboratory scheduling. Furthermore, technicians must manually search through handwritten logbooks to identify which computers are available or in need of repair. These findings highlighted critical procedural bottlenecks that the proposed system aims to resolve.

Group Discussions: Collaborative sessions were held at the University of Gondar with various stakeholders, including lab technicians and lab coordinators. These discussions focused on identifying specific pain points in the current manual reporting and inventory management processes. This collaborative approach allowed for the harmonization of user expectations and helped define the core functional requirements necessary for the new digital system.

2.2.2 Business Rules

BR-1: Access to the computer laboratory shall be restricted only to authorized University of Gondar (UoG) students, instructors, and lab administrators.

BR-2: Each user shall have an assigned role, and their activities shall be limited according to manual supervision by lab staff.

BR-3: Lab reservations shall be made only for registered students or staff who submit a written request to the lab assistant.

BR-4: All laboratory activities, including workstation usage and lab sessions, must be recorded in the lab logbook by authorized personnel.

BR-5: Equipment usage and status shall be updated only by authorized lab staff using

manual inventory registers.

BR-6: Records of student lab attendance and activities shall be maintained manually by lab personnel.

BR-7: Faults or damages to laboratory equipment shall be reported verbally or in written form to the lab technician.

BR-8: Maintenance and repair of laboratory equipment shall be scheduled and recorded manually by lab staff.

BR-9: Reports on lab usage, equipment status, and student performance shall be prepared manually when required.

BR-10: Notifications regarding lab availability or maintenance shall be communicated verbally or posted on notice boards.

BR-11: Students may request alternative lab access if a workstation or lab room is unavailable, following manual approval by the lab staff.

2.3 Proposed System Description

The proposed Computer Laboratory Management System (CLMS) will be a comprehensive digital system aimed at integrating all four university campuses of the University of Gondar. The new digital system will replace the traditional record-keeping approach employed in the university, thus integrating all four campuses of the University of Gondar for effective management of the resources, among other factors. Based on the main functions of the proposed Computer Laboratory Management System, the new system will comprise the UoG User Management and Authentication Module, which will allow for the safe access of the Computer Labs of the UoG Community while verifying the roles of students, instructors, and technicians in the institution.

The system also utilizes the University of Gondar Laboratory Scheduling and Bookings Module for the digitization of lab timetables across all campuses of UoG, thus allowing instructors to book rooms, but preventing session overlaps. To ensure there is technical accountability, the UoG Inventory and Asset Tracking Module holds a digital register of all hardware resources on site; thus, it enables the UoG technicians to know the physical location and specifications of each asset. In addition, the UoG Maintenance and Fault Reporting Module provides a direct communication line between users and

the maintenance team using an internal ticketing system, ensuring that all repairs must be recorded first before making any workstation as operational. Finally, the UoG Reporting and Analytics Module creates automated electronic reports for university management regarding the usage of labs and equipment health conditions, thus promoting transparency and informed decision-making for future ICT investments at the University of Gondar.

2.3.1 Overview

The proposed Computer Laboratory Management System project for the University of Gondar will significantly improve the management and utilization of the university's laboratories. The project involves developing a computer-based system to manage all administrative aspects for both students and the laboratories, including student attendance, laboratory allocation, inventory, and maintenance. With the help of a secure central database, the CLMS will provide a real-time solution for monitoring workstation utilization and availability across all the University of Gondar laboratories, marking a significant enhancement and modernization of the current manual record- keeping methods.

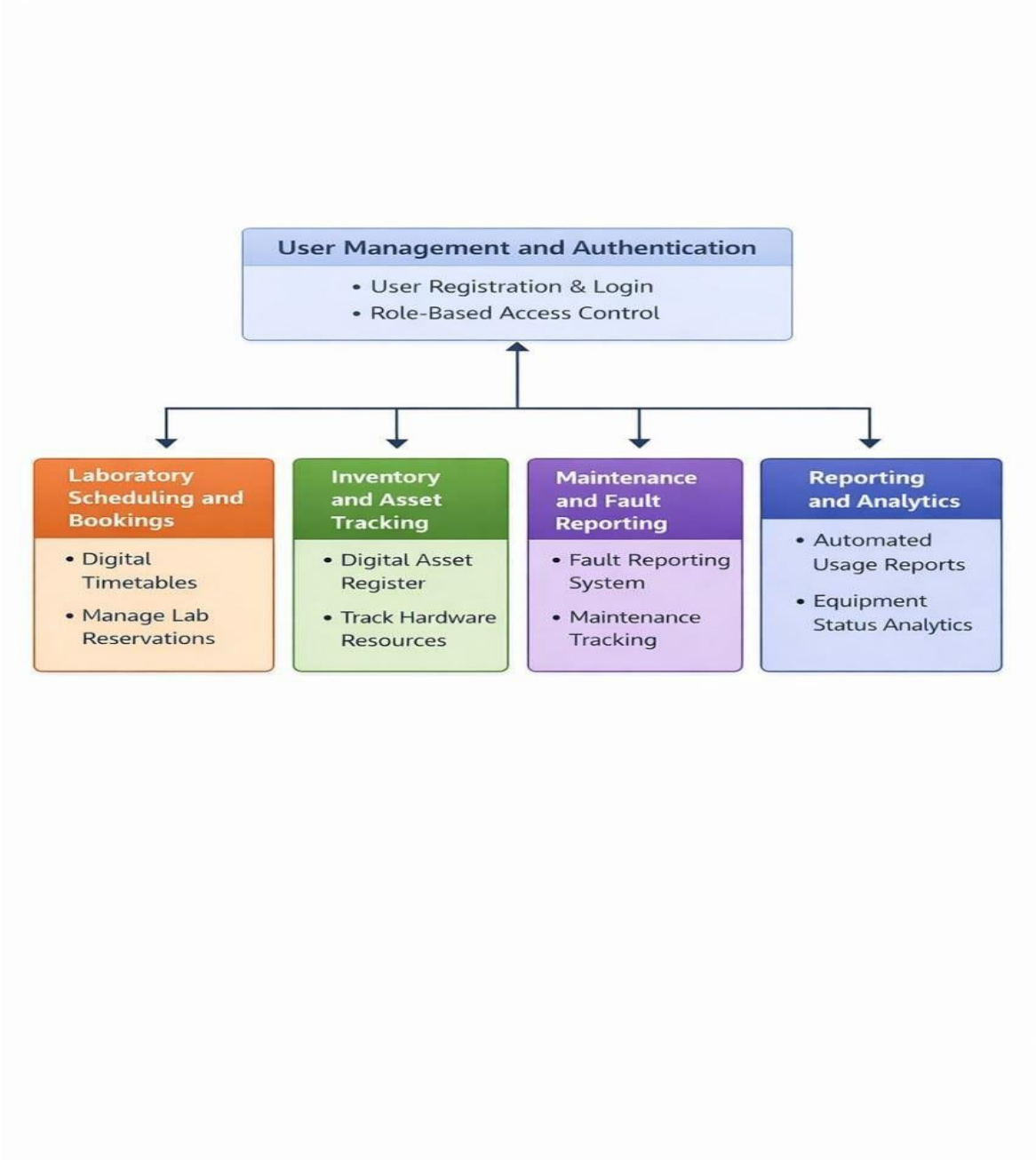

Figure 2.1: overview of the proposed system diagram

2.3.2 Functional Requirements

The functional requirements of the University of Gondar (UoG) CLMS are structured into specialized components to ensure seamless interaction across the university's four campuses.

The Student component is managed by granting authenticated students specific privileges, such as the ability to request workstation bookings and download lab manuals or materials. The independent flow for students follows a sequence of Login, View Availability, Booking Request, Download Materials, and Fault Reporting. In coordination, the Teacher component empowers faculty members to manage academic timetables and reserve laboratories. The teacher's flow involves Lab Reservation, Uploading Materials, View Lab Schedule, Fault Reporting, and Monitoring Student Attendance. Information is fully integrated between these modules: Student attendance and booking data flow to teachers, while technical tickets from both users flow to Technicians. In return, equipment status updates flow from Technicians to teachers to ensure only functional resources are scheduled.

For the sake of preserving the functionality and integrity of the hardware, the Lab Technician component allows technical staff to keep abreast of the status of the workstations and act on student-submitted help tickets. The status of the equipment can be updated in real time, indicating which is active, faulty, and being repaired, while keeping an updated digital record of the entire complement of hardware and software. Administrative management is provided through an organizational structure that has the Campus Administrator monitoring daily activity, scheduling, and staff performance for a designated campus site, while the Super Administrator has overall strategic power for the entire multi-campus system. The Super Admin has system-level configuration and Analytical University of Gondar-Wide reporting, and Management of User-Role Rights at the system-wide and strategic institutional administrative supervision and management.

component	Roles	Key Functionalities	Workflow
Student component	Students	Request workstation bookings, download lab materials/manuals, Report faults, and receive notifications.	Login, View Availability, Booking Request, Download Materials, Fault Reporting; Receives alerts for bookings and maintenance
Teacher component	Teachers	Manage academic timetables, Reserve labs, upload teaching materials, monitor student attendance, and receive equipment status updates.	Lab Reservation, Upload Materials, Fault Reporting, View Lab Schedule, Monitor Attendance; Integration with student bookings and technician updates
Lab Technician component	Technicians	Monitor workstation status, respond to help tickets, Update equipment status in real-time (Active, Faulty, Under Repair), maintain digital inventory of hardware/software, and generate maintenance reminders.	Tracks all lab resources, View Lab Schedule, updates ticket status, handles fault reporting, and ensures equipment integrity.
Administrator component	Admin	Monitor daily lab activities, Schedule lab sessions, Track staff performance, and approve bookings.	Manages operations and staff performance for the individual campus
Super Administrator component	Super Admin	University-wide system configuration, Analytical reporting, managing user roles and permissions, Multi-campus supervision, and Managing Admin Account.	Strategic management, overall monitoring, and reporting across all campuses

Table 2.1: Summary of functional requirements

2.3.3 Non- Functional Requirements

Non-functional requirements specify the CLMS's performance objectives, constraints, and quality attributes. The following parameters are recommended and measured to guarantee that the system continues to be sustainable and effective within the academic setting of the University of Gondar:

2.3.3.1 Performance

Response time and throughput are used to gauge the system's performance. It must respond to all of the main user modules with efficiency. For example, the system needs to react in two to three seconds when a student looks up the availability of a workstation or an administrator creates a report. For handling high requests during peak usage, the system employs background processing of requests as well as indexing of databases based on key fields to enable quick data retrieval without refreshing entire pages.

2.3.3.2 Scalability

Concurrent User Capacity and Data Volume Capacity are used to gauge the CLMS. It must grow with the University of Gondar without experiencing a decline in performance. Both vertical hardware upgrades and functional expansions are made possible by support for a growing concurrent user base (up to 1,000 simultaneous users) across the Maraki, Atse Tewodros, Fasil, and GC campuses.

2.3.3.3 Availability

The System Uptime Percentage measures availability. The system aims for 99.9% uptime, ensuring laboratory services can be accessed. In addition, inspired by advanced reservation systems, say, airline bookings, if all workstations in a given lab are reserved, the system will automatically display No Workstations Available and suggest other times or nearby available labs to the client. Alerts are sent to the Super Admin on the unavailability of services.

2.3.3.4 Reliability

Mean Time to Failure (MTTF) and Mean Time to Recovery (MTTR) are ways of measuring reliability. The system ensures that its operations are continuous by allowing fault tolerance and making sure data is handled securely. With automated checks and backup procedures, user information and lab schedules are protected in such a way that in case the system fails, a recovery would take less than 1 hr.

2.3.3.5 Maintainability

Because it was designed, from the ground up, to be maintainable, the system is modular, with pieces that can be updated independently without taking down the whole system. This would include updates in the Technician's ticketing interface. Heavy documentation and standardized coding practices support rapid troubleshooting and new functionality in the future, ensuring a long-term, sustainable software investment.

2.3.3.6 Security

The measures of security are based on the access violation rate and encryption coverage. The access mechanism uses role-based access control. The data integrity mechanism uses the 100% encryption of sensitive credentials, which also monitors for malicious access in order to minimize the risk of unauthorized entry and ensure the highest level of system integrity.

2.3.3.7 Environmental

Apart from digitizing business processes with an attempt to eliminate paper usage, this system caters to User Comfort and Ergonomics. The design ensures minimal eye fatigue (through proper contrast and scaling of fonts), and a comfort zone will thus be achieved in this system with a digital interface. Later automation of power management will give Technicians a capability to put unused machines into energy-saving modes.

2.3.3.8 Usability

Usability is measured by Task Completion Time and Error Rates. The interface is intuitive, allowing students and faculty to perform tasks such as creating a booking or generating a report in less than 3 minutes. The logical workflow minimizes the number of clicks to perform key tasks, making the system highly accessible without extensive training.

2.3.3.9 Interoperability

The CLMS is designed in a way that allows for a smooth flow of data between its different modules, such as User Management, Inventory, and Maintenance. The system allows for a smooth transition where data from one module can be instantly accessible in another module in order to maintain a smooth flow of operations. For example, in case a fault is reported in the system under the Maintenance and Fault Reporting module, the condition of the equipment will automatically be reflected in the Inventory and Asset Tracking module. This allows all modules within the system to be in sync, such that reports in the Reporting and Analytics module will be based on real-time data from all other modules within the system.

CHAPTER THREE: SYSTEM MODEL

3.1 Scenario

As part of our analysis for the proposed Computer Laboratory Management System (CLMS), we have developed detailed usage scenarios to clearly illustrate the problems with the current manual system and how the new digital platform will transform laboratory operations at the University of Gondar. These scenarios involve all key stakeholders and demonstrate the tangible benefits of our proposed solution. To help you visualize the transformation, I've merged the current pain points with the proposed digital solutions for each stakeholder. This comparison highlights how the Computer Lab Management System (CLMS) shifts the experience from manual frustration to automated efficiency.

1. The Student Scenario

The Student transitions from a physical wanderer to a digital planner. Currently, they must walk across campuses (Maraki, Atse Tewodros, Fasil, and Gc.) only to find full labs or broken equipment, followed by the hassle of manual logbooks. In the proposed system, the student checks real-time availability from their dorm, secures a seat via a web portal, and checks in instantly with a QR code. Instead of discovering a broken mouse too late, they can report faults via their phone with a photo, and the system immediately redirects them to a working workstation.

2. The Teacher Scenario

For the Teacher, the system replaces administrative red tape with one-click scheduling. The current reality involves physical forms and the frequent double-booking nightmare where two classes collide. The proposed CLMS provides a unified view of all campus labs, preventing scheduling conflicts automatically. Beyond booking, the teacher evolves into a digital facilitator, uploading materials, taking automated QR-code attendance, and reviewing post-session analytics to see how students utilized the lab time.

3. The Lab Technician Scenario

The Lab Technician shifts from data entry clerk to technical specialist. Instead of spending hours deciphering handwritten logbooks and manually compiling Excel reports, the system automates all tracking and documentation. Maintenance requests no longer get lost in paper notebooks; they are assigned as digital tickets that the assistant can update in real-time. This eliminates lost requests and ensures that equipment downtime is minimized through a prioritized digital workflow.

4. The Administrator Scenario

The Administrator moves from guesswork to data-driven precision. Currently, procurement and budgeting are based on vague estimates because there is no way to see real-time lab health. With the CLMS dashboard, the administrator gains instant visibility into occupancy rates and equipment longevity. This transparency ensures that the budget is spent where it’s actually needed, preventing unnecessary expenditures and ensuring resources are redistributed to the busiest labs.

5. The Super Admin Scenario

The Super Admin gains centralized oversight across a fragmented geography. Currently, there is no way to monitor security or usage across Maraki, Atse Tewodros, Fasil, and GC simultaneously. The proposed system provides a God's-eye view of all campuses from a single portal. The Director can now manage permissions, receive security alerts for hardware tampering, and use predictive maintenance insights to plan long-term institutional growth rather than just reacting to daily crises.

3.1.1 Use Case Model

Based on the provided document and the use case diagram, the Use Case Model for the University of Gondar Computer Laboratory Management System (CLMS) is presented as follows:

Actor	Description	Key Responsibilities
Student	A registered university student uses the computer lab for coursework and projects.	Users can register an account, view lab and workstation availability, book a workstation, download learning materials, and report faulty equipment.

Teacher	A lecturer conducts and manages lab-based teaching sessions.	Users can reserve entire lab rooms, upload course materials, view the lab schedule, and monitor student attendance.
Lab Technician	Technical support staffs maintain the labs and assist during sessions.	View Lab Schedule, update the repair status of faulty equipment, and perform regular equipment status checks.
Admin	The manager is responsible for daily operations and user management for a specific department.	Approve new user registrations, manage all user accounts, generate usage and inventory reports, manage system inventory, and detect and resolve booking conflicts.
Super Admin	The system director has full control over the entire CLMS across all campuses.	View system audit logs, manage all user roles and permissions, and configure all core system settings.

Table 3: Actor Specification

3.1.2 Use Case Diagram

A Use Case Diagram is a unified modeling language (UML) that provides a visual representation of the functional requirements of a system from the user's point of view. In essence, it provides a model of how the system will be used by different actors. In the context of the Computer Laboratory Management System (CLMS) for the University of Gondar, the Use Case Diagram will help to identify the different actors in the system, including Students, Lab Assistants, Technicians, and Administrators. In addition, it will provide a clear representation of the goals that can be achieved in the system. For instance, the system will be used to register for lab sessions, book workstations, report faults in the system, keep track of inventory, and generate reports. In essence, the Use Case Diagram will provide a clear blueprint that ensures all the

needs of the stakeholders are catered to. The needs will be identified through document analysis, observation, and group discussions.

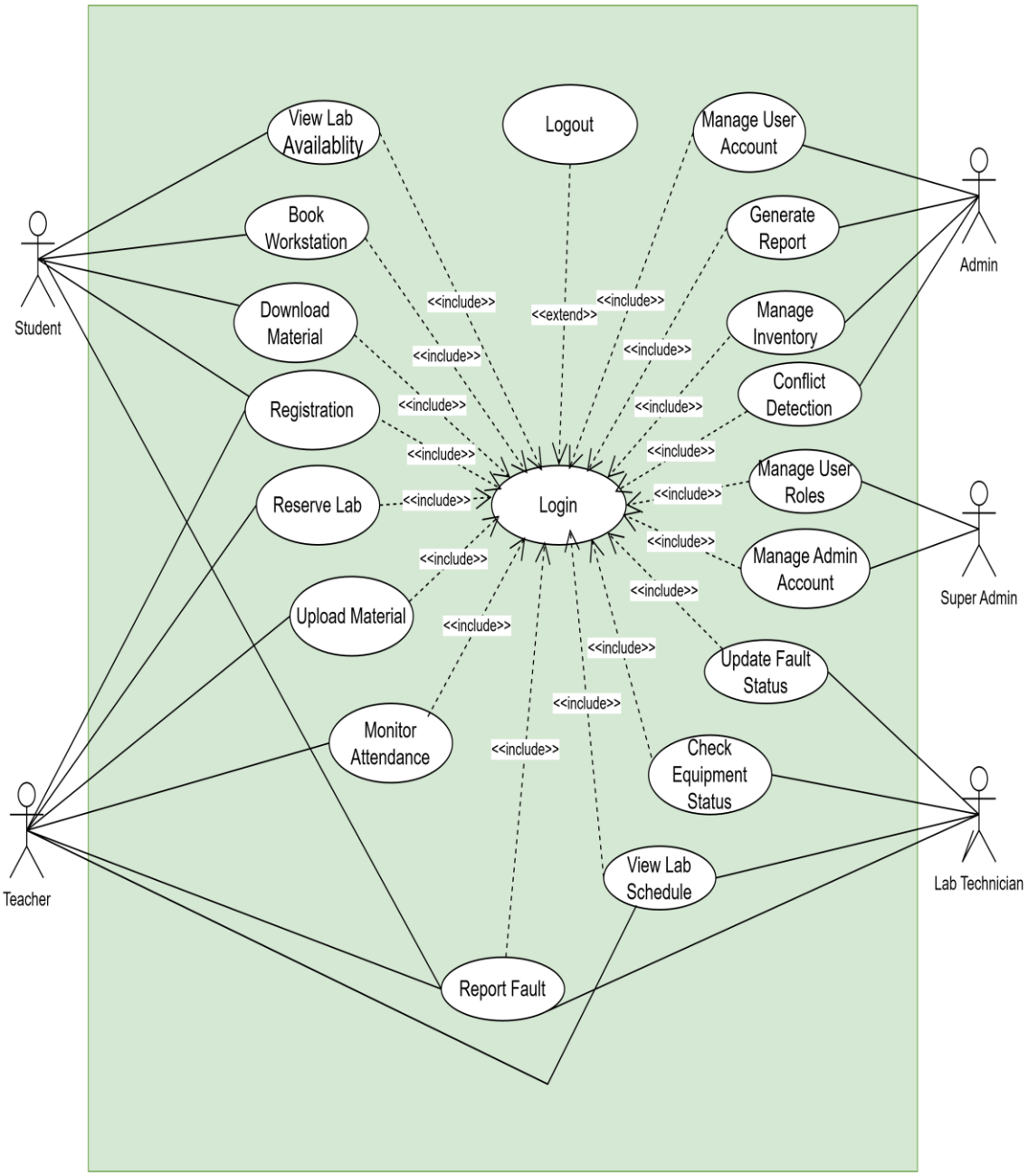

Figure 3.1: Use Case Diagram

3.1.3 Use Case Descriptions

Use Case ID	UC-01	
Use Case Name	Login	
Actors	Student, Teacher, Lab Technician, Administrator, Super Admin	
Pre-condition	The user must have a registered account and be on the home page.	
Basic Flow of Events	User Action	System Action
	1. User selects the Login link.	2. System displays the login form.
	3. User enters username and password.	4. System validates the authenticity of the credentials.
	5. User clicks the Login button.	6. System grants access and displays the user-specific dashboard.
Alternative Flow	If the username or password is invalid:1. The system displays Incorrect username or password. 2. System clears the password field and stays on the login page.	
Post-condition	The user is successfully authenticated and logged into the system dashboard.	

Table 3.1: Description of Login

Use Case ID	UC-02	
Use Case Name	Book Workstation	
Actor	Student	
Pre-condition	The student must be logged into the system.	
Basic Flow of Events	User Action	System Action
	1. Student selects View Lab Availability.	2. System displays a real-time map of all PCs and their status.
	3. Student selects an available workstation.	4. System verifies the PC is not Faulty.
	5. Student clicks Book Now.	6. System confirms the booking and generates a QR code for check-in.
Alternative Flow	A1: Workstation Unavailable If the selected PC is marked as Faulty or taken during the process: 1. System displays Workstation is currently unavailable. 2. System refreshes the map and suggests another available seat.	
Post-condition	The workstation is successfully reserved for the student.	

Table 3.2: Description of Book Workstation

Use Case ID	UC-03	
Use Case Name	Reserve Lab	
Actor	Teacher	
Pre-condition	The teacher must be logged into the system.	
Basic Flow of Events	User Action	System Action
	1. Teacher selects the Reserve Lab menu.	2. System displays the reservation form (Date, Time, and Room).
	3. Teacher enters session details.	4. System performs Conflict Detection check.
	5. Teacher clicks the Submit button.	6. System confirms reservation and sends notification.
Alternative Flow	A1: Schedule Conflict If the lab is already reserved for the same time slot: 1. System displays Conflict Detected: Lab is already booked. 2. The system shows a list of alternative available time slots or rooms.	
Post-condition	The lab room is reserved, and the schedule is updated.	

Table 3.3: Description of Reserve Lab

Use Case ID	UC-04	
Use Case Name	Report Fault	
Actors	Student, Lab Technician, and Teacher	
Pre-condition	The user must be logged into the system.	
Basic Flow of Events	User Action	System Action
	1. User selects Report Fault link.	2. System displays the fault report form.
	3. User enters PC number and issue description.	4. System validates the workstation ID.
	5. User clicks the Submit button.	6. System updates PC status to Faulty and notifies the Technician.
Alternative Flow	A1: Invalid Workstation ID If the PC number entered does not exist in the inventory: 1. System displays "Invalid PC Number, please try again". 2. System stays on the report form.	
Post-condition	The fault is logged, and the maintenance team is notified.	

Table 3.4: Description of Report Fault

Use Case ID	UC-05	
Use Case Name	Manage Inventory	
Actor	Administrator	
Pre-condition	The administrator must be logged into the system.	
Basic Flow of Events	User Action	System Action
	1. Admin selects the Manage Inventory menu.	2. System displays the current hardware list.
	3. Admin selects Add New.	4. System displays the data entry form.
	5. Admin fills in hardware details and clicks Save.	6. System saves the changes and displays Inventory Updated Successfully.
Alternative Flow	A1: Duplicate Entry If the hardware serial number already exists: 1. System displays Error: Serial number already exists. 2. The system asks the Admin to verify the information.	
Post-condition	The hardware inventory list is successfully updated.	

Table 3.5: Description of Manage Inventory

Use Case ID	UC-6	
Use Case Name	Conflict Detection	
Actor	Admin	
Pre-condition	Admin must be logged in; triggered during lab registration or booking.	
Basic Flow of Events	User Action	System Action
	1. Admin selects Conflict Detection or attempts to approve a schedule.	2. System scans the database for overlapping time slots and lab rooms.
	4. Admin reviews the highlighted conflicts.	3. System identifies any double-bookings or resource overlaps.
		5. System suggests alternative free time slots.
Alternative Flow	A1: No Conflicts Found. System displays No scheduling conflicts detected.	
Post-condition	Scheduling conflicts are identified and resolved.	

Table 3.6: Description of Conflict Detection

Use Case ID	UC-7	
Use Case Name	Manage User Account	
Actors	Admin	
Pre-condition	Admin must be logged in.	
Basic Flow of Events	User Action	System Action
	1. Admin selects Manage User Account.	2. System displays a list of registered users and search options.
	3. Admin chooses to Update, Disable, or Create an account.	4. System displays the relevant user information form.
	5. Admin submits the changes.	6. System updates the database and confirms Account Updated Successfully.
Alternative Flow	A1: Invalid User ID System displays User not found if the searched ID is incorrect.	
Post-condition	User account details (Student, Teacher, and Technician) are created, updated, or disabled.	

Table 3. 7: Description of Manage User Account

Use Case ID	UC-8	
Use Case Name	Generate Report	
Actors	Admin	
Pre-condition	Admin must be logged into the system.	
Basic Flow of Events	User Action	System Action
	1. Admin selects the General Report menu from the dashboard.	2. System displays various report types (e.g., Monthly usage, Inventory status, Fault history).
	3. Admin selects the desired report type and specifies the date range.	4. The system gathers relevant data from the database and generates the report.
	5. Admin clicks on the Download or View button.	6. System displays the report in a readable format.
Alternative Flow	A1: No Data Available If there is no data for the selected criteria: 1. System displays No records found for the selected period. 2. System returns the Admin to the report selection screen.	
Post-condition	A comprehensive report on lab usage and performance is generated and displayed.	

Table 3.8: Description of the Generate Report

Use Case ID	UC-9	
Use Case Name	Monitor Attendance	
Actors	Teacher	
Pre-condition	Teacher must be logged in and a lab session must be active.	
Basic Flow of Events	User Action	System Action
	1. Teacher selects the Monitor Attendance menu.	2. System displays the list of students currently checked into the lab.
	3. Teacher verifies the list and clicks Save Attendance.	4. The system marks the students as present and saves the record.
Alternative Flow	A1: Manual Entry If a student's QR code fails: 1. Teacher manually searches for the student ID. 2. The teacher marks the student as present.	
Post-condition	Student attendance is digitally recorded and stored.	

Table 3.9: Description of Monitor Attendance

Use Case ID	UC-10	
Use Case Name	Download Material	
Actor	Student	
Pre-condition	The student must be logged in, and the teacher must have uploaded materials.	
Basic Flow of Events	User Action	System Action
	1. Student selects Download Material from the dashboard.	2. System displays a list of available files organized by course/teacher.
	3. The student selects the specific file and clicks Download.	4. System validates the file's presence and starts the transfer.
		5. System records the download in the student activity log.
Alternative Flow	A1: File Not Found If the file has been moved or deleted by the admin, the system displays File currently unavailable.	
Post-condition	The student successfully downloads the educational files.	

Table 3.10: Description of Download Material

Use Case ID	UC-11	
Use Case Name	Logout	
Actors	Student, Teacher, Lab Technician, Administrator, Super Admin	
Pre-condition	The user must be currently logged into the system.	
Basic Flow of Events	User Action	System Action
	1. User clicks the Logout button.	2. System clears the session data and cookies.
		3. System redirects the user to the home page or login screen.
Alternative Flow	A1: Auto-Logout If the user is inactive for a predefined period, the system automatically triggers the logout process for security.	
Post-condition	The session is securely terminated, and the user is redirected to the login page.	

Table 3.11: Description of Logout

Use Case ID	UC-12	
Use Case Name	Registration	
Actors	Student, teacher	
Pre-condition	The student must have a valid University ID.	
Basic Flow of Events	User Action	System Action
	1. User selects Registration from the home page.	2. System displays the registration form.
	3. User fills in personal and academic details and submits.	4. System validates that the Student ID is unique.
		5. System saves the data and notifies the Administrator for approval.
Alternative Flow	A1: Duplicate Registration If the ID is already registered, the system displays User already exists and suggests the Login option.	
Post-condition	A pending account is created and sent for Admin approval.	

Table 3.12: Description of Registration

ID	UC-13	
Use Case Name	Check Equipment Status	
Actor	Lab Technician	
Pre-condition	The Lab Technician must be logged into the system.	
Basic Flow of Events	User Action	System Action
	1. Lab Technician selects Check Equipment Status.	2. System displays a list of all workstations and their current health status.
	3. Technician filters by lab room or status Faulty.	4. System generates the filtered view.
Alternative Flow	A1: No Data Found 1. If the inventory is empty, the system displays No equipment records found.	
Post-condition	The current status of all lab equipment is reviewed.	

Table 3.13: Description of check equipment status

Use Case ID	UC-14	
Use Case Name	View Lab Availability	
Actor	Student	
Pre-condition	The student must be logged into the system.	
Basic Flow of Events	User Action	System Action
	1. Student selects the View Lab Availability link from the dashboard.	2. System retrieves the current status of all labs and workstations from the database.
	3. The student selects a specific lab room to check.	4. System displays a real-time graphical map showing which PCs are Available, booked, or Faulty.
Alternative Flow	A1: Lab Closed/Full 1. If the lab is currently closed for maintenance or fully occupied, the system displays a Lab Unavailable message. 2. System suggests checking other available lab rooms.	
Post-condition	The student is able to see the real-time status of the laboratory and individual workstations.	

Table 3.14: Description of View Lab Availability

Use Case ID	UC-15	
Use Case Name	Upload Material	
Actor	Teacher	
Pre-condition	The teacher must be logged into the system and have appropriate permissions.	
Basic Flow of Events	User Action	System Action
	1. Teacher selects the Upload Material option from the dashboard.	2. System prompts for file selection and metadata (title, description, course).
	3. Teacher chooses the file and enters metadata.	4. System validates the file format and size.
	5. Teacher confirms upload.	6. System stores the file in the repository and updates the course material list.
Alternative Flow	A1: Invalid Fill 1. If the file format or size is invalid, the system displays an error message. 2. System suggests acceptable formats and size limits.	
Post-condition	The material is successfully uploaded, stored in the database, and accessible to students.	

Table 3.15: Description of Upload Material

3.1.4 Activity Diagram

Activity diagrams are used in the documentation of the logic involved in the specific component of the CLMS, such as the Lab Booking module, the Inventory Management module, and the Maintenance Reporting module. They include the drawing of flowcharts that show the flow of control between activities within the modules. Activity diagrams also model the sequential processes involved in the functional modules of the system. In addition, they represent the flow of an object from one state to another at various points in the flow of control.

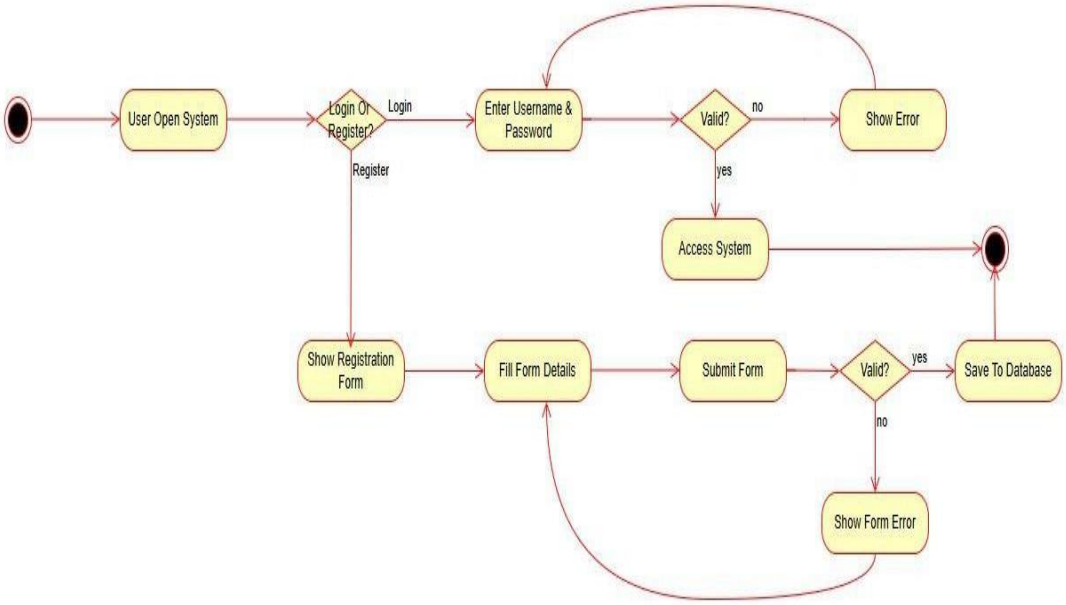

Figure 3.2: Activity Diagram of User Management

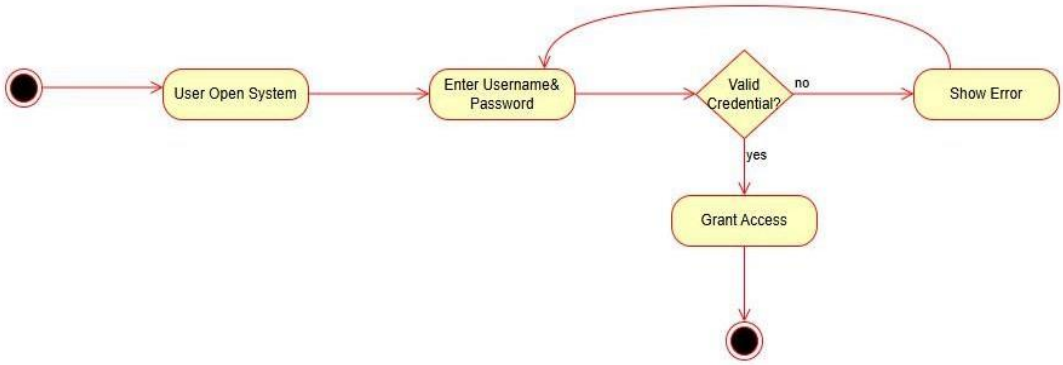

Figure 3.3: Activity Diagram of Authentication

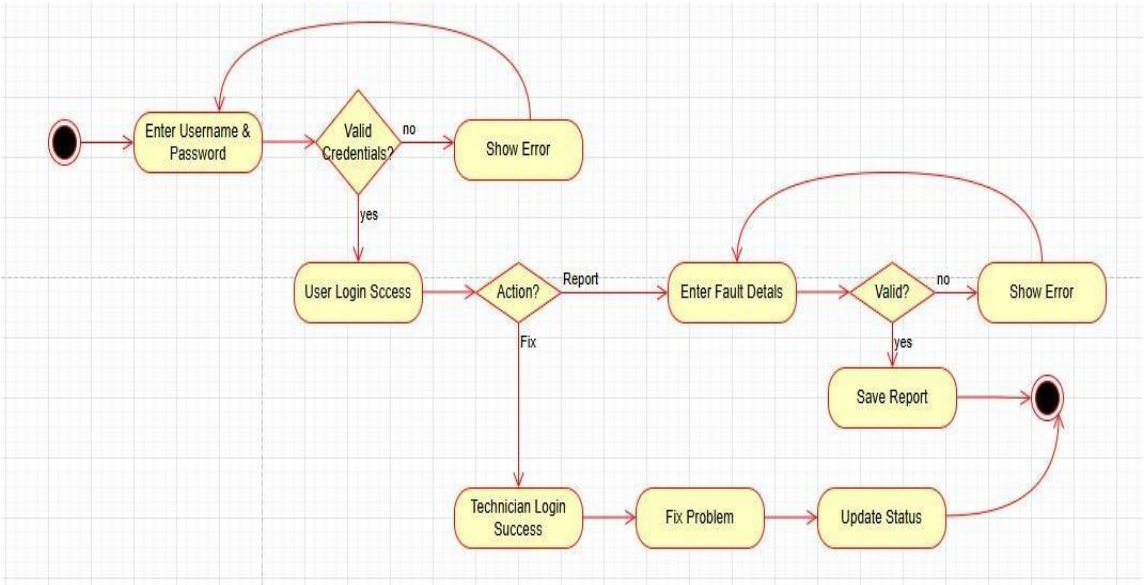

Figure 3.4: Activity Diagram of Maintenance

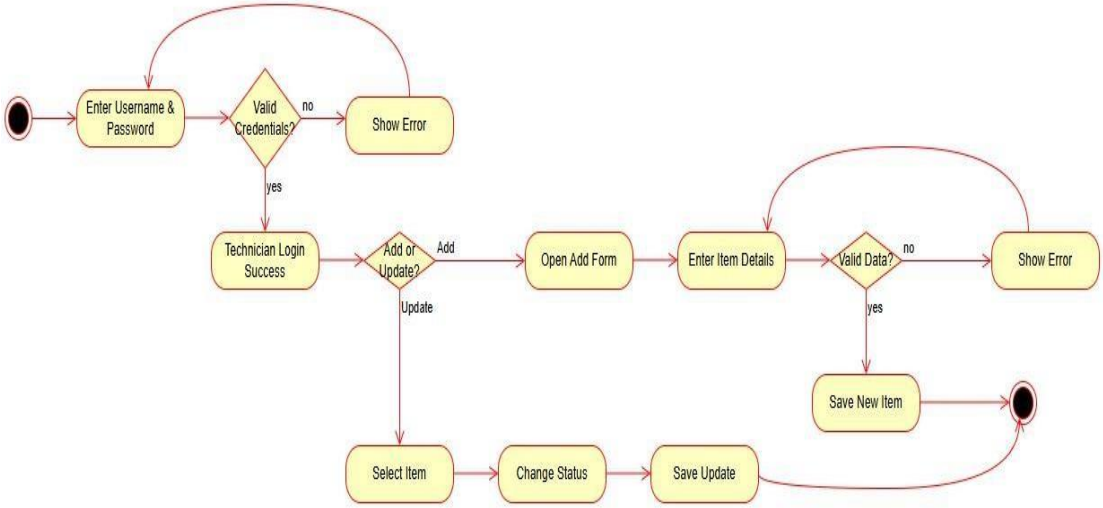

Figure 3. 5: Activity Diagram of Inventory Tracking

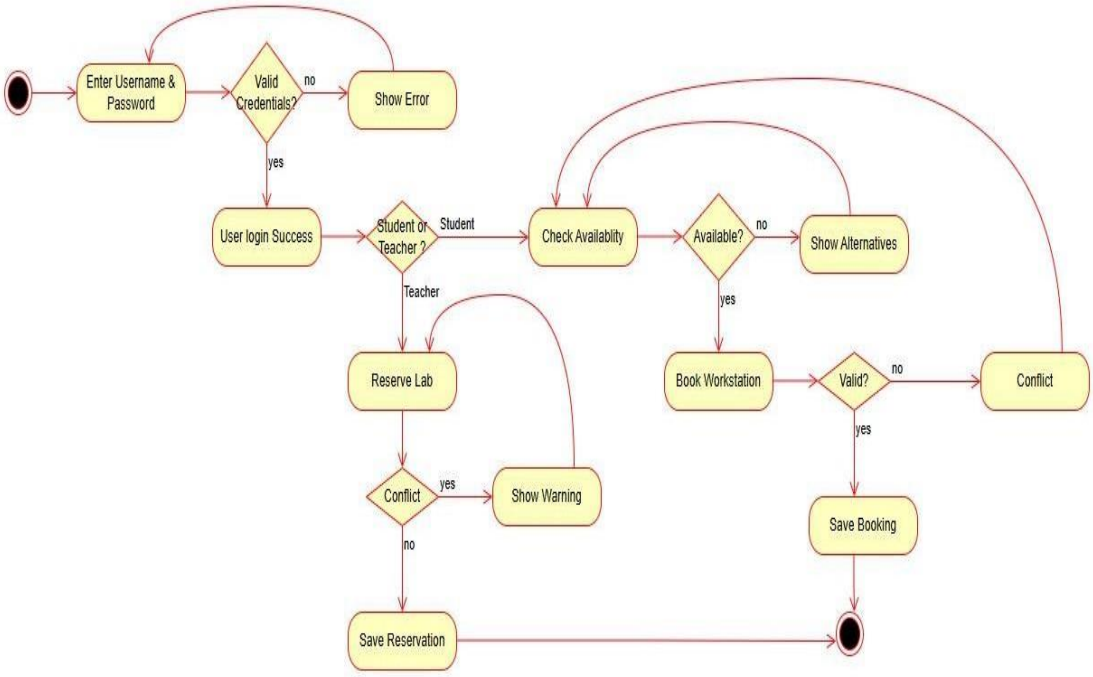

Figure 3. 6: Activity Diagram of Lab Booking

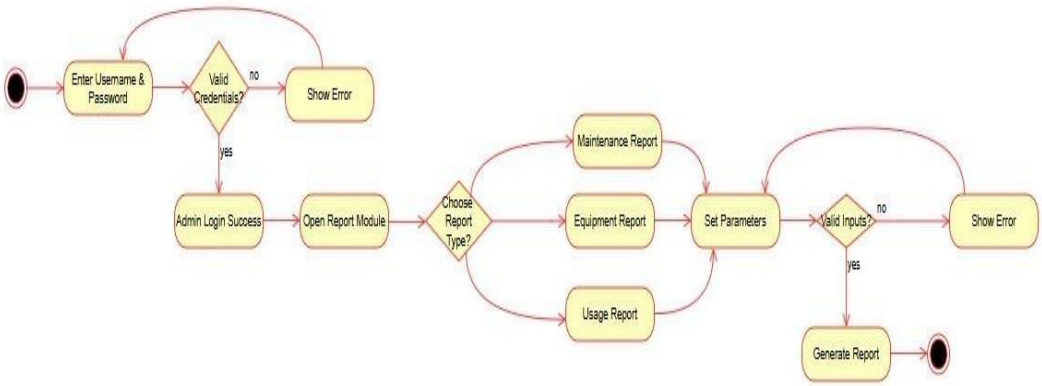

Figure 3.7: Activity Diagram of Reporting

3.1.5 Object Diagram

The Object Diagram shows the main entities in the Computer Laboratory Management System and how they connect. It illustrates real-world objects used in managing computer laboratories at the University of Gondar, along with their attributes and states. Each object includes its unique identifier, relevant attributes, and links to other objects. This diagram offers a clear view of the system's data and acts as a basis for database design and class modeling.

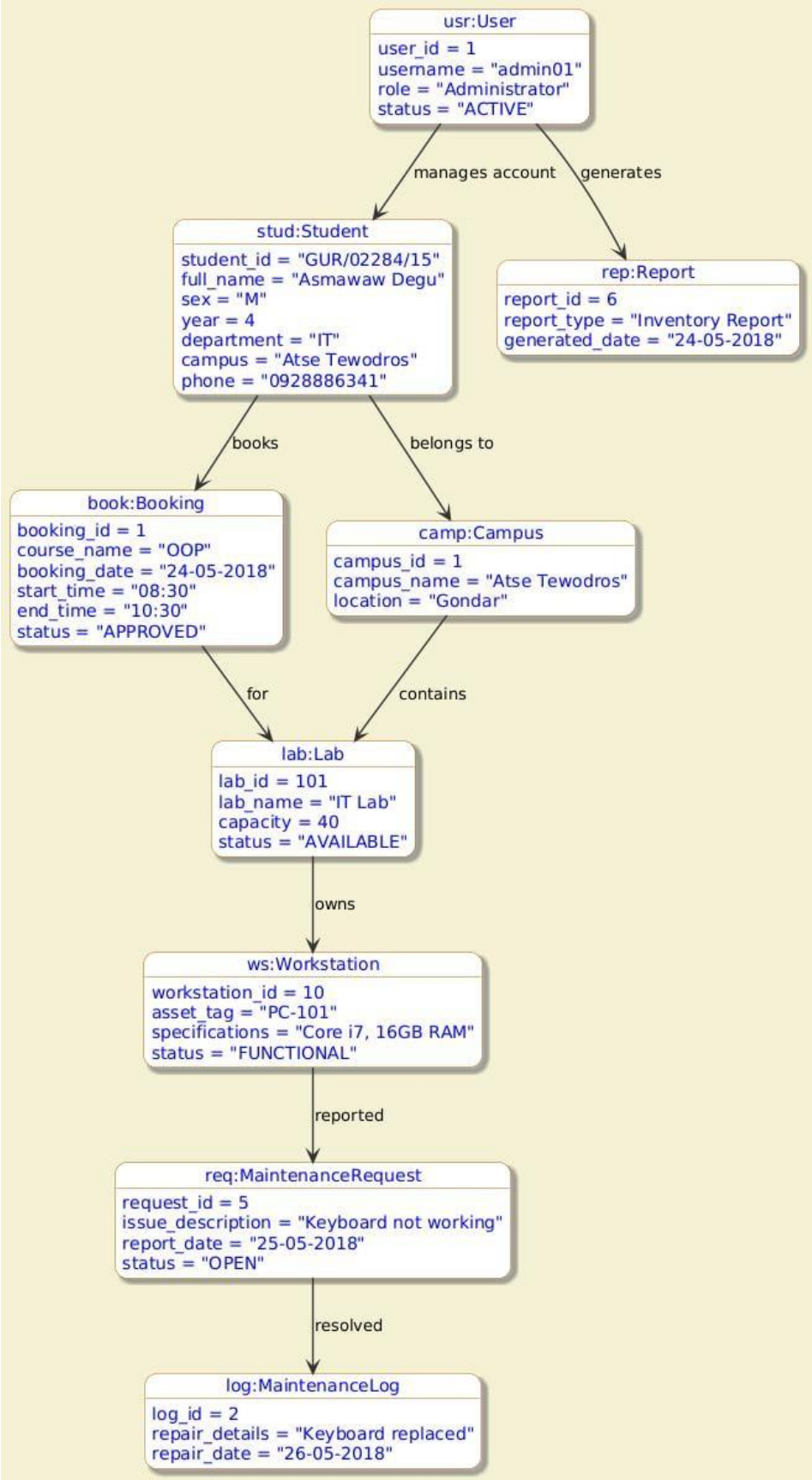

Figure 3.8**:** Object Diagram

3.1.6 Data Dictionary

Data Item	Class	Data Type	Data Format	Storage Size	Display Size	Description	Example	Validation
Username	User	String	Alphanumeric	30	30	User login name	admin01	Must be unique
Password	User	String	Encrypted	255	12	User password	******	Minimum 6 characters
User Role	User	String	Admin / Instructor / Technician / Student	20	20	System user role	Administrator	String only
User Status	User	String	ACTIVE / INACTIVE	10	10	Account status	ACTIVE	String only
Student ID	Student	String	Alphanumeric	20	20	Unique student ID	GUR/02284/15	Must be unique
First Name	Student	String	Text	30	30	Student first name	Asmamaw	Alphabetic only
Last Name	Student	String	Text	30	30	Student last name	Degu	Alphabetic only
Sex	Student	String	MALE / FEMALE	6	6	Student gender	MALE	String only

Department	Student	String	Text	50	50	Student department	IT	Not null
Campus	Campus	String	Text	30	30	Campus name	Atse Tewodros	Predefined
Phone Number	Student	String	Numeric	10	10	Contact number	0928886341	Must be 10 digits
Lab Name	Laboratory	String	Text	50	50	Laboratory name	IT Lab	Not null
Lab Capacity	Laboratory	Integer	Numeric	4	4	Number of computers	40	Must be > 0
Workstation Status	Workstation	String	Functional / Faulty / Under Repair	20	20	Workstation condition	Functional	String only
Booking Date	Booking	Date	YYYY-MM-DD	10	10	Date of booking	2026-02-01	Valid date
Start Time	Booking	Time	HH: MM	5	5	Session start time	08:30	Valid time
End Time	Booking	Time	HH:MM	5	5	Session end time	10:30	Valid time
Booking Status	Booking	String	PENDING / APPROVED /	15	15	Booking status	APPROVED	String only

			CANCELLED					
Issue Description	Maintenance Request	String	Text	200	100	Maintenance issue details	Keyboard not working	Not null
Maintenance Status	Maintenance Request	String	OPEN / IN PROGRESS / RESOLVED	20	20	Maintenance progress	OPEN	String only
Report Type	Report	String	Text	30	30	Report category	Inventory Report	Not null

Table 3.16: Data Dictionary

3.1.7 Class Model

The class model represents the static structure of the Web-Based Computer Laboratory Management System (CLMS). It illustrates the key classes in the system, along with their attributes, methods, and the relationships among them, including inheritance, association, and aggregation. Each class in the model represents a real-world entity that participates in the management of computer laboratories at the University of Gondar.

3.1.7.1 Class Diagram

The class diagram is one of the most significant diagrams in UML. It is used to model the static structure of a system. The class diagram models classes, their properties, and methods, as well as the relationships between objects such as inheritance, association, and aggregation. In the Web-Based Computer Laboratory Management System (CLMS), the class diagram shows how users such as Super Admin, Campus Admin, Instructor, Student, and Technician interact with system entities, including Campus, Laboratory, Workstation, Booking, Ticket, Inventory, Maintenance Log, and Report. The class diagram provides a basis for system design, implementation, and database creation.

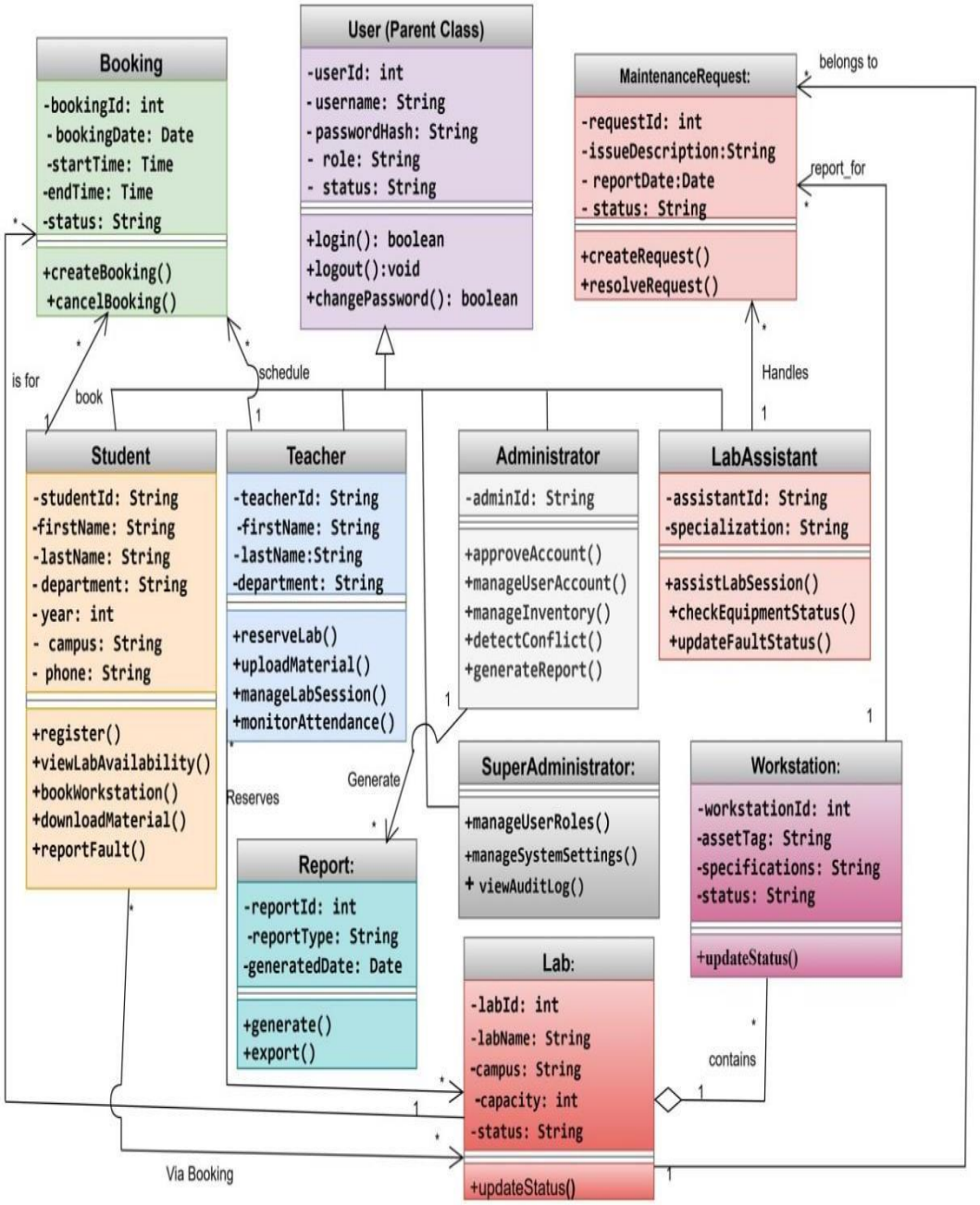

Figure 3.9: Class Diagram

3.1.8 Dynamic Modeling

Dynamic modeling provides an understanding of how the Web-Based Computer Laboratory Management System (CLMS) behaves over time through an explanation of how objects interact with each other, how messages are passed between objects, and how their states change during operation. It focuses more on system behavior rather than structure and is developed based on the identified entity, boundary, and control objects along with their attributes and associations.

Dynamic modeling can be achieved through Sequence Diagrams and State Chart Diagrams, which provide an understanding of system workflows and object life cycles.

3.1.8.1 Sequence Diagram

A UML sequence diagram represents the sequence of interactions between objects and is used to represent or model the flow of messages, events, and actions between the objects or components of a system, illustrating how different system objects communicate or interact with each other over time in order to achieve a particular task or scenario. In the context of the Web-Based Computer Laboratory Management System (CLMS), sequence diagrams are used to represent interactions involving user login, lab booking, workstation usage, fault reporting, maintenance handling, and report generation, and they support the designing, documenting, and validating of the system architecture and interfaces by illustrating the sequence of actions that must be performed to accomplish a specific task within the system.

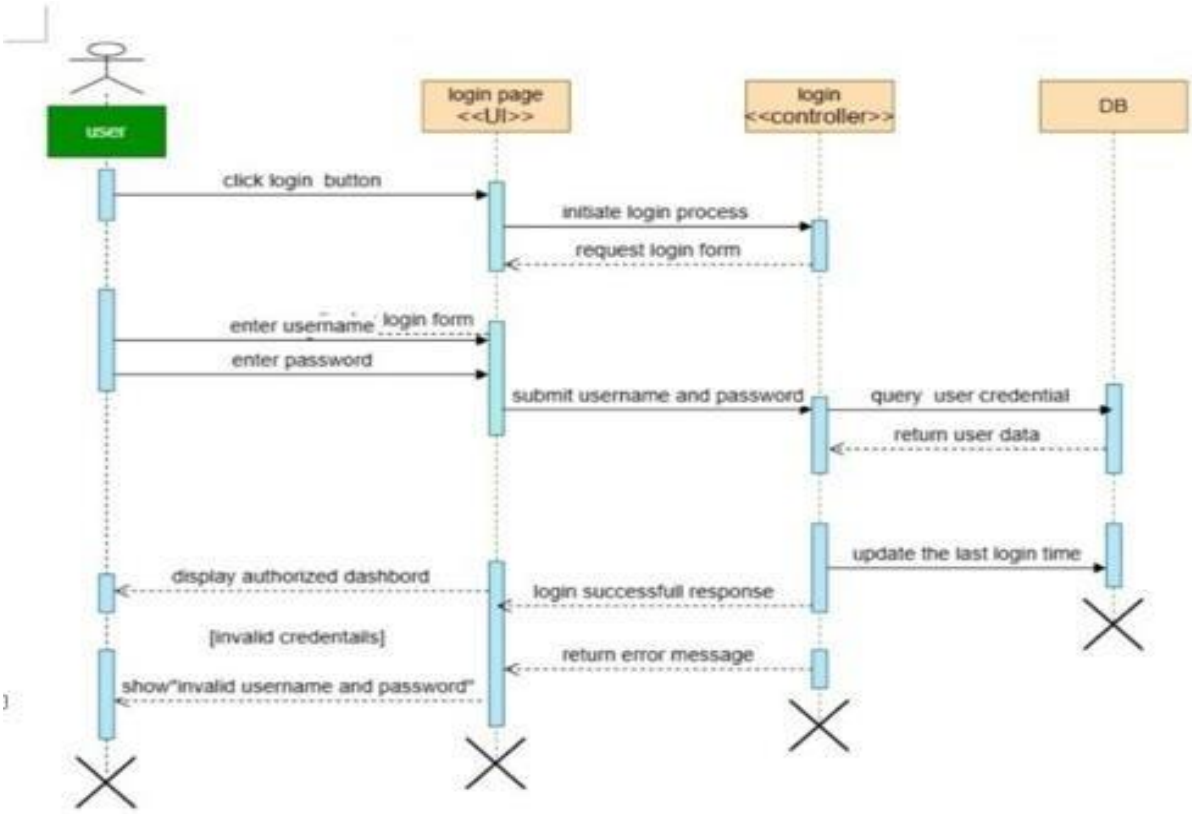

Figure 3.10: Login Sequence diagram

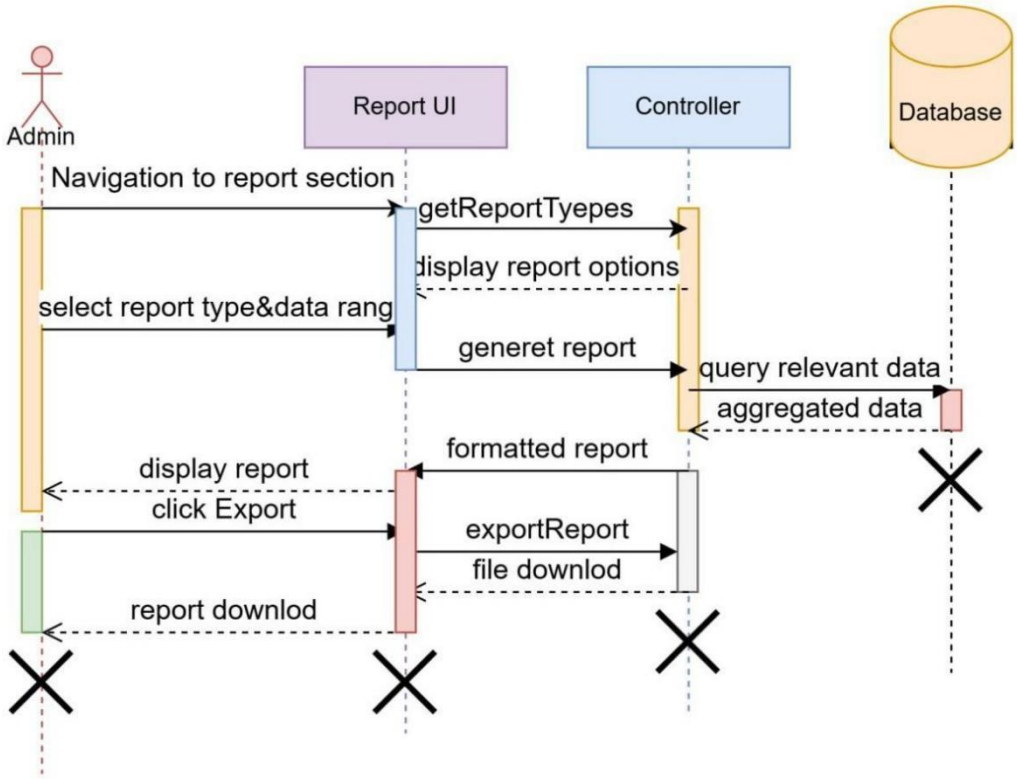

Figure 3.11: Report Sequence diagram

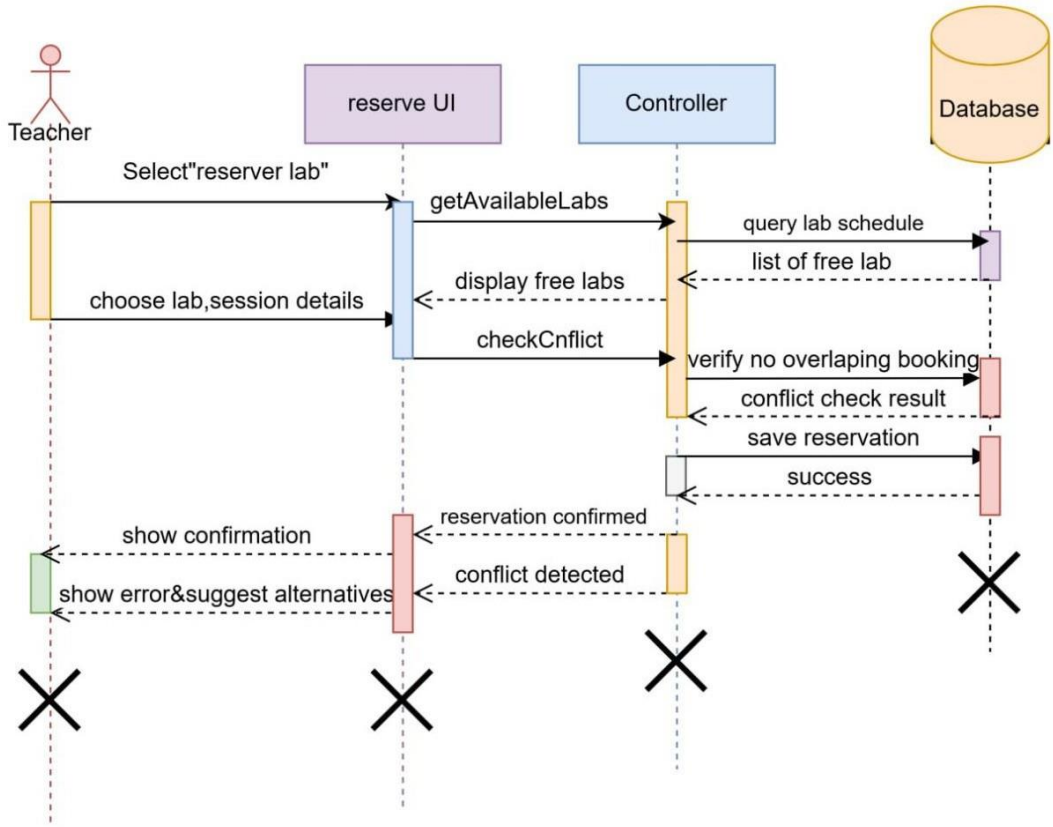

Figure 3.12: Reserve Sequence diagram

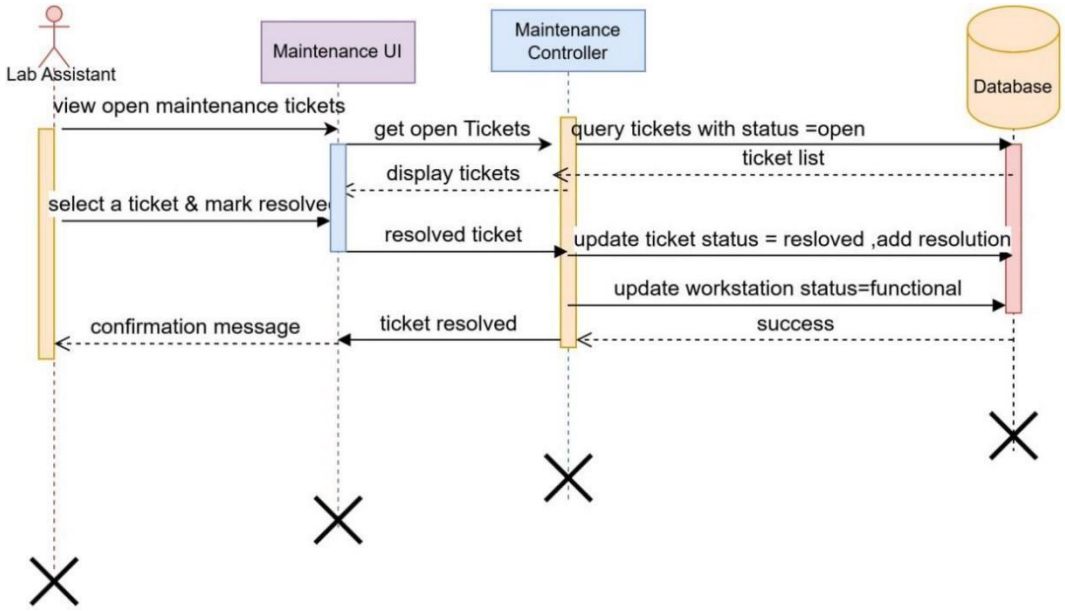

Figure 3.13: Maintenance Sequence diagram

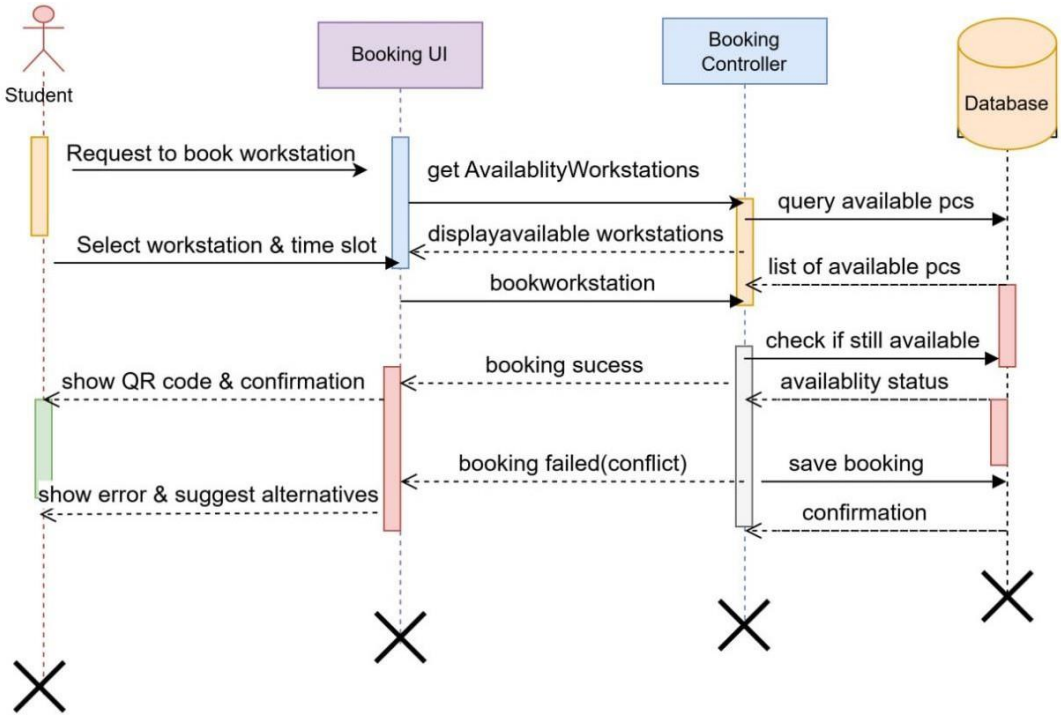

Figure 3.14**:** Booking Sequence diagram

3.1.8.2 State Chart Diagram

The state chart diagram represents the dynamic behavior of an object by showing the different states that an object goes through during its lifetime. The events that occur during transitions from one state to another, as well as the actions that are performed during these transitions, are also represented.

In the Web-Based Computer Laboratory Management System (CLMS), state chart diagrams are used for booking, workstation, and maintenance ticket objects. For example, a booking object may have different states such as requested, approved, scheduled, completed, or canceled. Similarly, a maintenance ticket object may have states such as reported, in progress, and resolved. The state chart diagram ensures that object behavior occurs as required by the defined business rules.

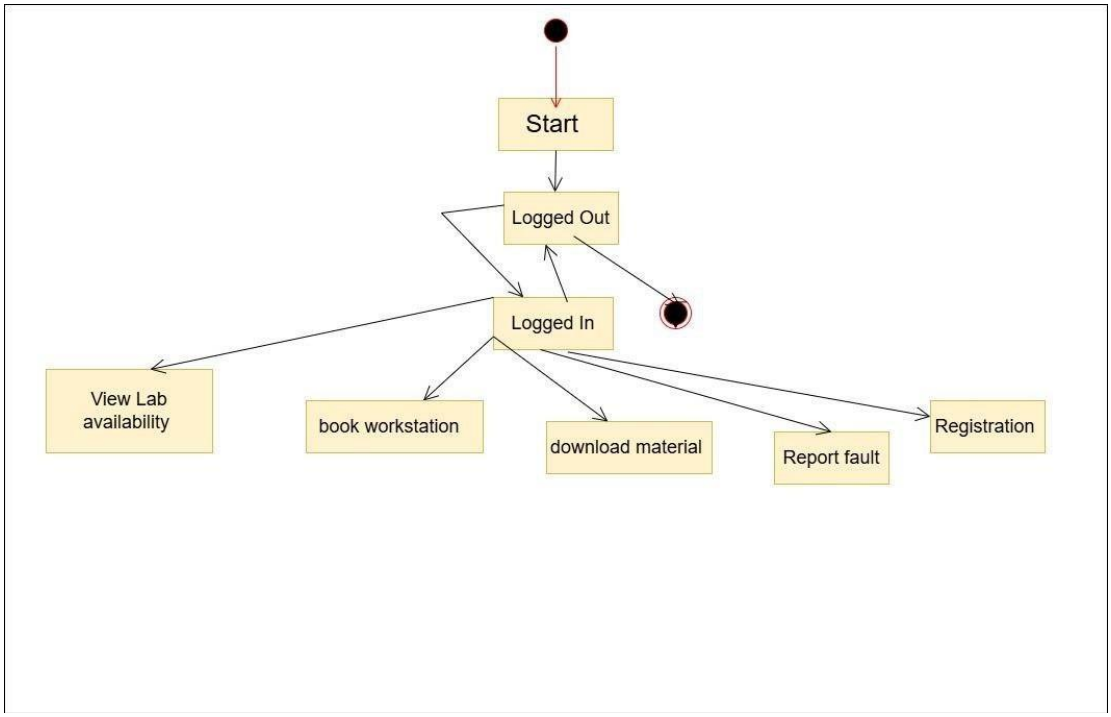

Figure 3.15: State Chart Diagram of Student

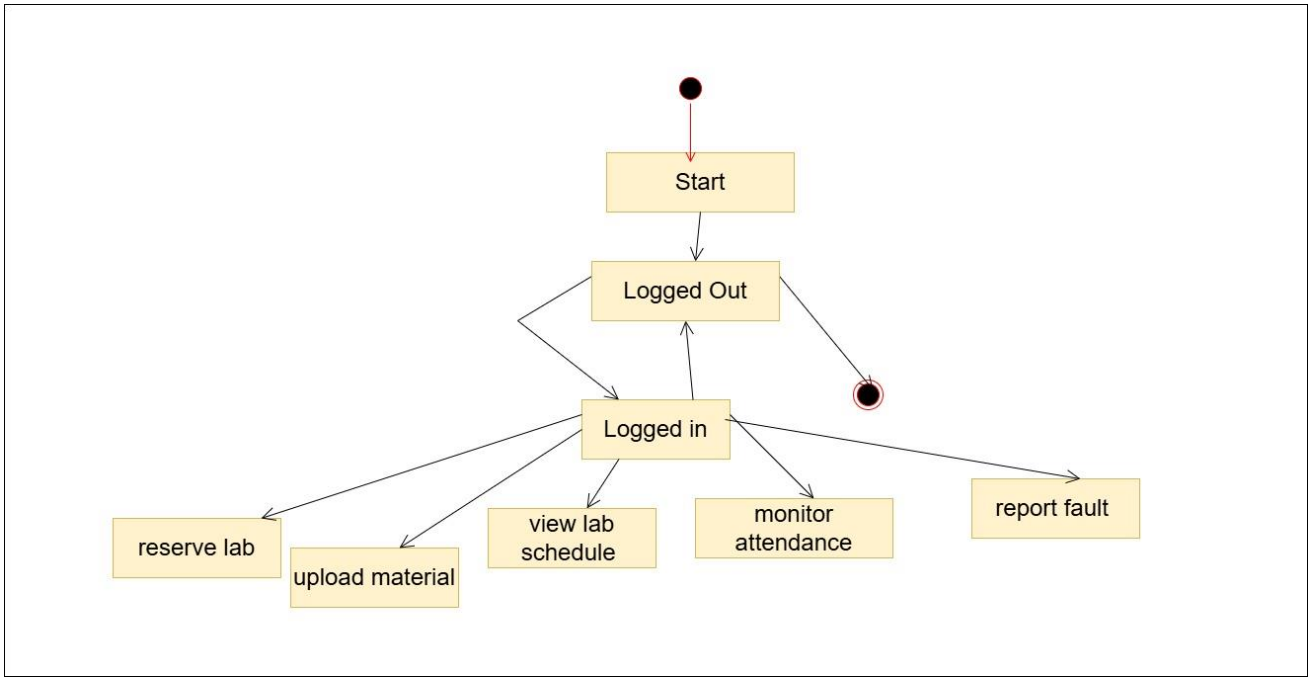

Figure 3.16: State Chart Diagram of Teacher

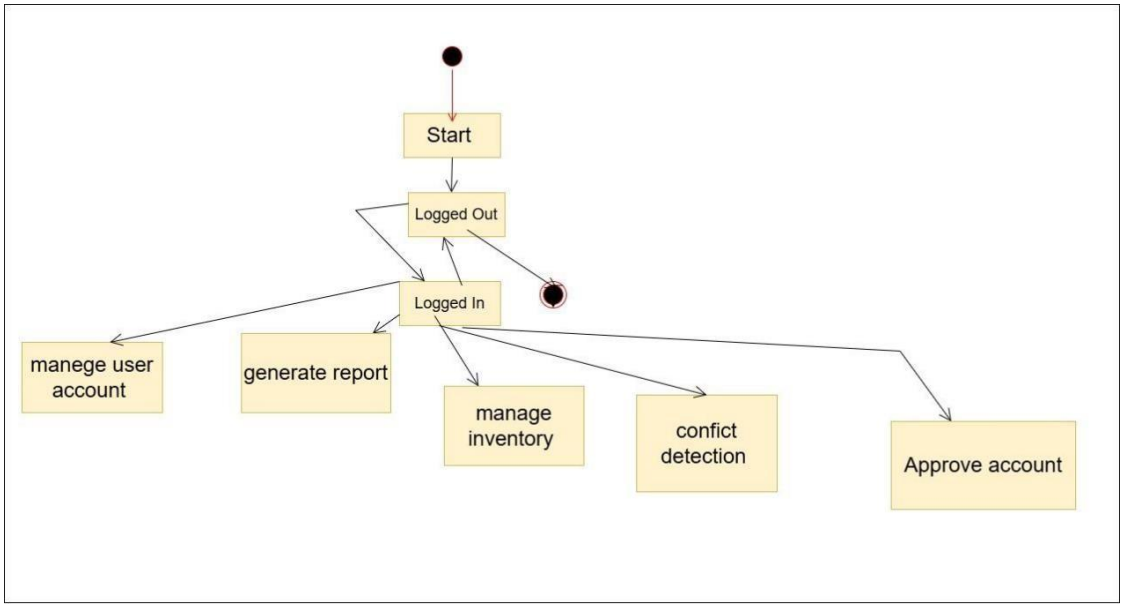

Figure 3.17: State Chart Diagram of Admin

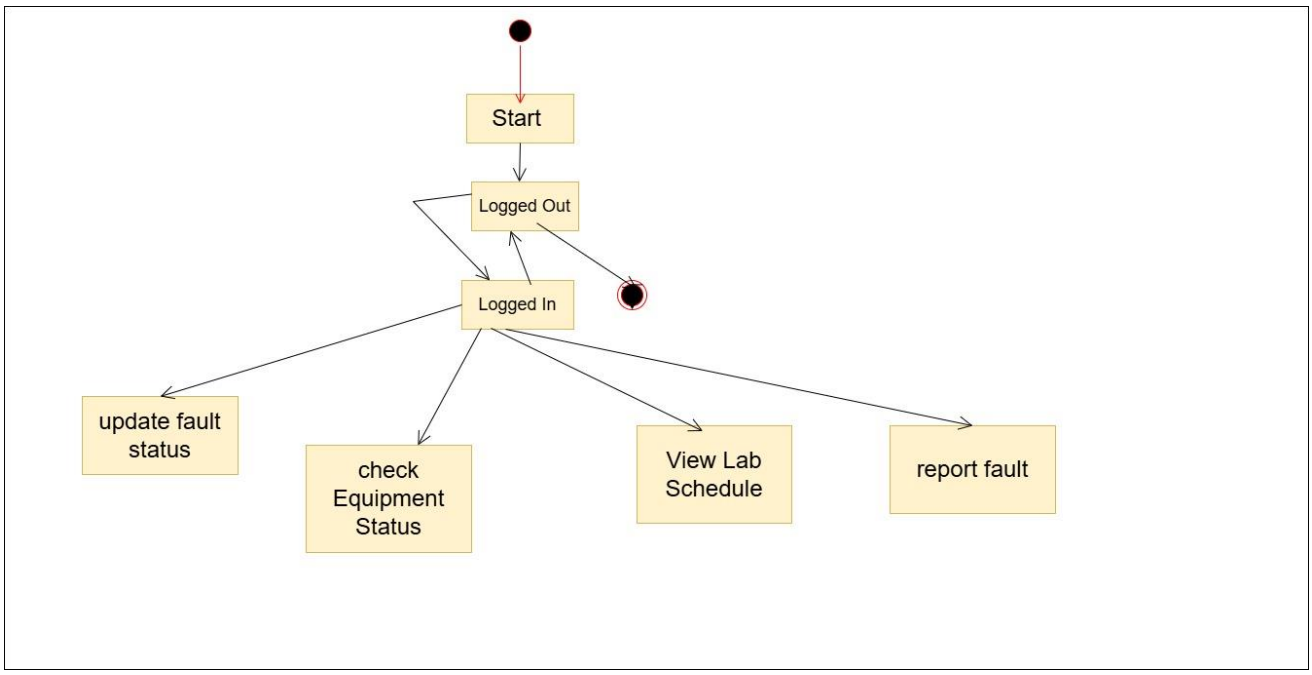

Figure 3.18: State Chart Diagram of a Lab Technician

3.1.9 User Interface Prototype

This section presents the user interface design of the Web-Based Computer Laboratory Management System (CLMS) through screen mock-ups, navigational paths, and interface specifications. The design follows user-centered principles to ensure usability,

accessibility, and efficiency for all stakeholders, including students, instructors, lab technicians, and administrators.

Figure 3.19: Login/Register User Interface

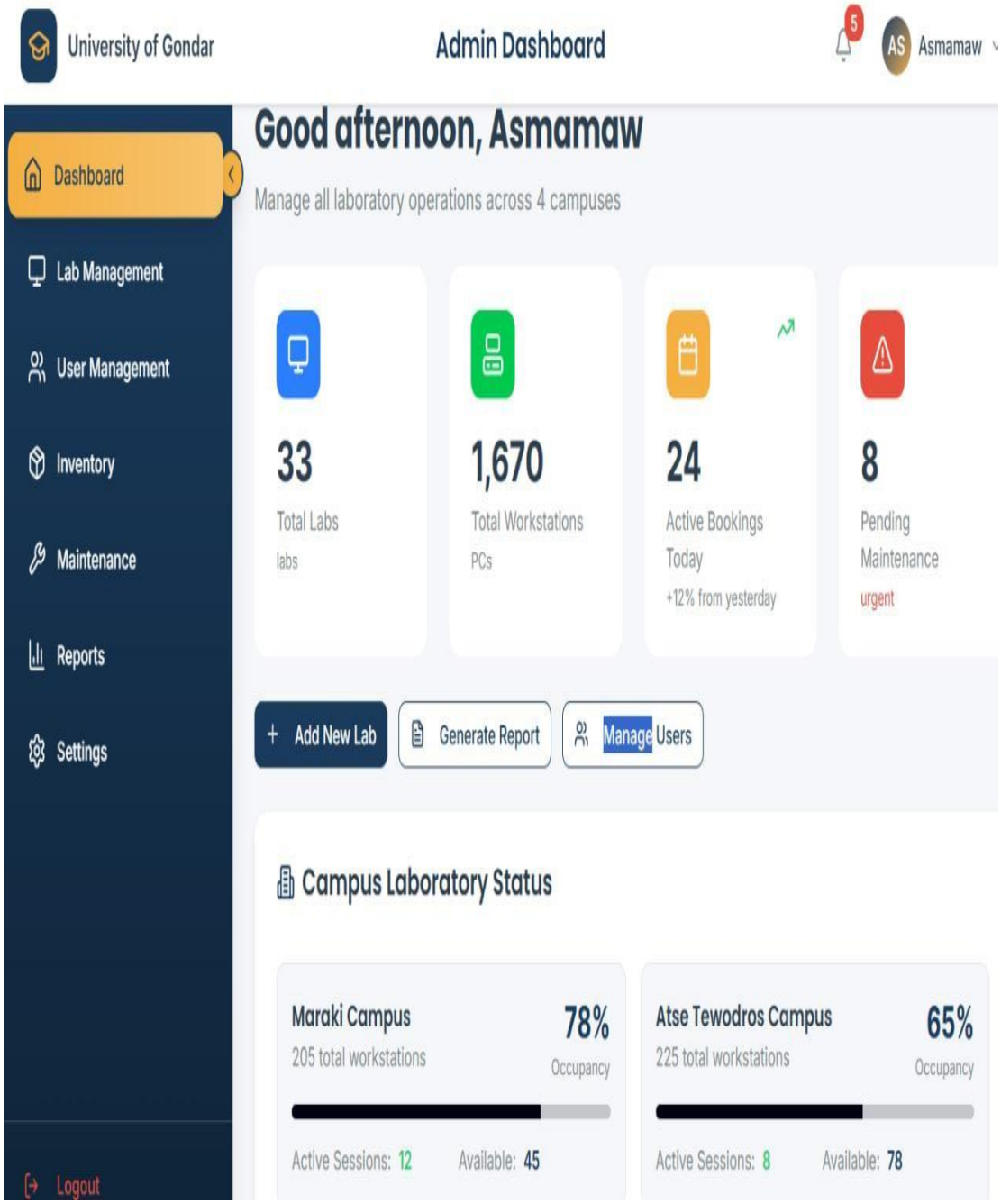

Figure 3.20: Admin Dashboard User Interface

CHAPTER FOUR: SYSTEM DESIGN

4.1 Introduction

In the previous chapter, we discussed the functional and non-functional requirements of the Computer Laboratory Management System (CLMS**)** and produced the analysis model. This chapter focuses on transforming that analysis model into a formal design model that takes into account the non-functional requirements and constraints, such as data security and concurrent user access, described in the problem statement and requirement analysis sections discussed earlier. We provide an overview of the current manual laboratory handling and the proposed CLMS software architecture, while specifying design goals to improve the quality of the system by reducing the response time for workstation bookings, increasing the availability of lab schedules across different campuses (Maraki, Atse Tewodros, Fasil, and GC), and improving its reliability to function correctly under erroneous conditions. Software architecture for the CLMS is the process of defining a structured solution that meets all technical and operational requirements while optimizing common quality attributes such as performance, security, and manageability. We describe the hardware/software mapping with a deployment diagram, subsystem decomposition, access control, and security protocols. Hardware/software mapping in this system involves mapping CLMS subsystems to web servers and database components, enabling us to identify potential concurrency among subsystems and to address performance and reliability goals. We also specify the boundary conditions for each subsystem by mentioning how the modules are initialized and terminated. This document is strictly traceable to the previous Software Project Management Plan (SPMP) and Software Requirement Specification (SRS) documents.

4.2 Current Software Architecture

Since the existing Computer Laboratory Management System at the University of Gondar is a manual and paper-based system, there is no formal software architecture currently in place. The current architecture consists of physical storage such as paper logbooks, handwritten reservation forms, and maintenance notebooks scattered across various campus administrative offices. Because the system is not supported by a digital framework, it lacks essential architectural components such as a central server, a

database, or application logic. Consequently, the information flow is entirely physical, requiring students and teachers to be present on-site to access services or verify workstation availability, which results in a complete lack of real-time data synchronization or digital connectivity between the different campuses.

As illustrated in the diagram, the current manual system begins with different users, including students, teachers, and lab assistants. These users provide information through manual input**,** which is recorded using logbooks and paper forms. After data entry, the information is processed manually by lab staff through human processing**,** where tasks such as laboratory scheduling and equipment inventory management are performed. The processed information is then stored in physical storage, such as files and record books. Finally, the results or updates are communicated through outputs like notice boards and public announcements. This process clearly demonstrates that the current system relies entirely on manual operations, which increases the chances of delays, data loss, and inefficiencies in laboratory management.

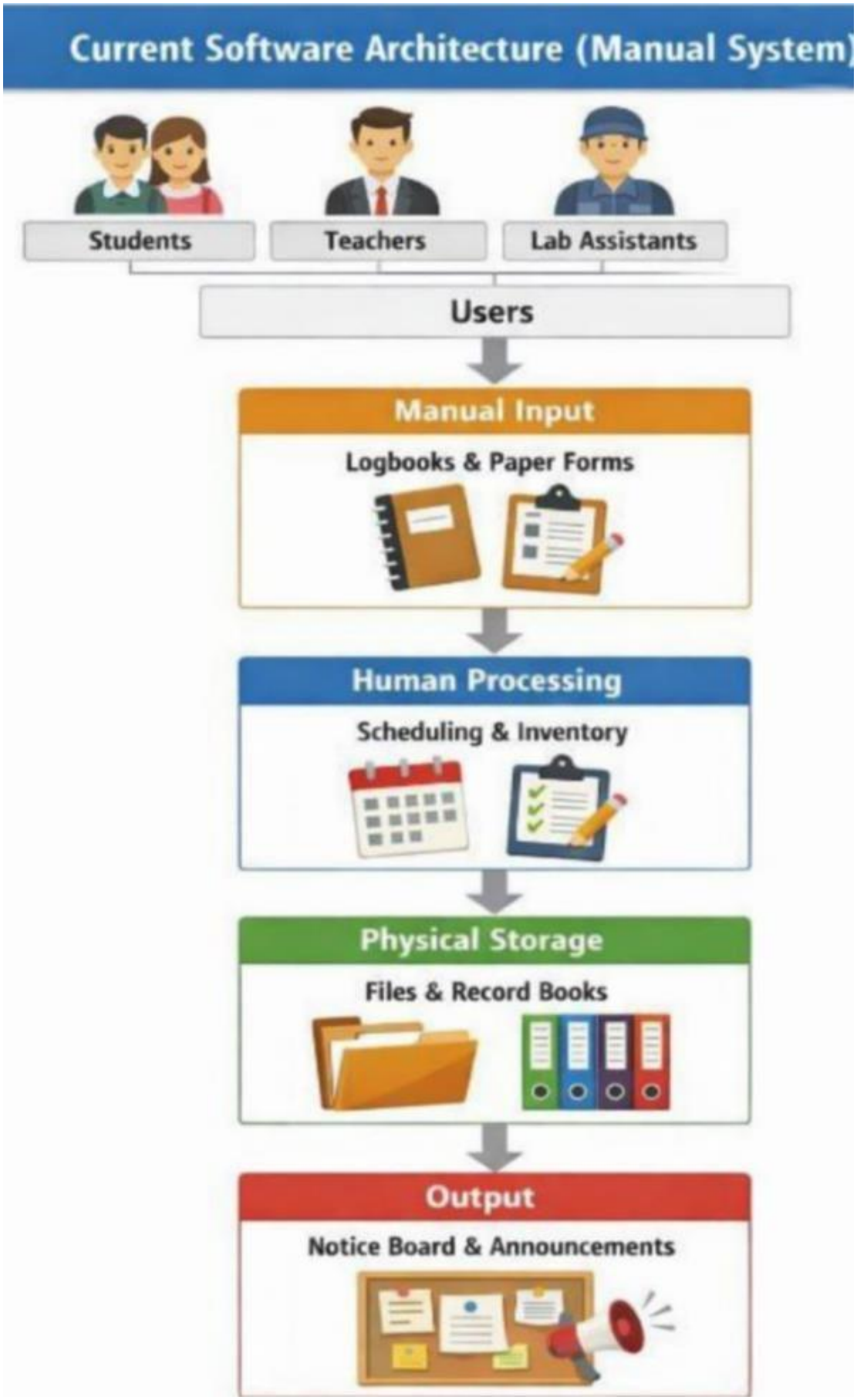
Current Software Architecture (Manual System)
Students
Teachers
Lab Assistants
Users
Manual Input
Logbooks & Paper Forms
Human Processing
Scheduling & Inventory
Physical Storage
Files & Record Books
Output
Notice Board & Announcements

Figure 4.1: Current Software Architecture

4.3 Proposed Software Architecture

The proposed CLMS adopts a web-based, multi-tier client-server architecture designed to centralize laboratory management and provide high availability across all university campuses. This architecture is divided into three distinct layers: the Presentation Layer, the Application Logic Layer, and the Data Layer. The Presentation Layer provides the web portal through which students, teachers, and administrators interact with the system via responsive dashboards. The Application Logic Layer serves as the core of the system, residing on a centralized web server to process business rules such as automated booking validations, maintenance ticketing workflows, and user authentication. Finally, the Data Layer consists of a centralized Relational Database Management System (RDBMS) that ensures persistent and secure storage of all university records, hardware inventory, and lab schedules. By adopting this tiered structure, the system ensures that a fault reported in the maintenance module is immediately reflected in the inventory and booking status, replacing the fragmented manual process with a synchronized, real-time management environment.

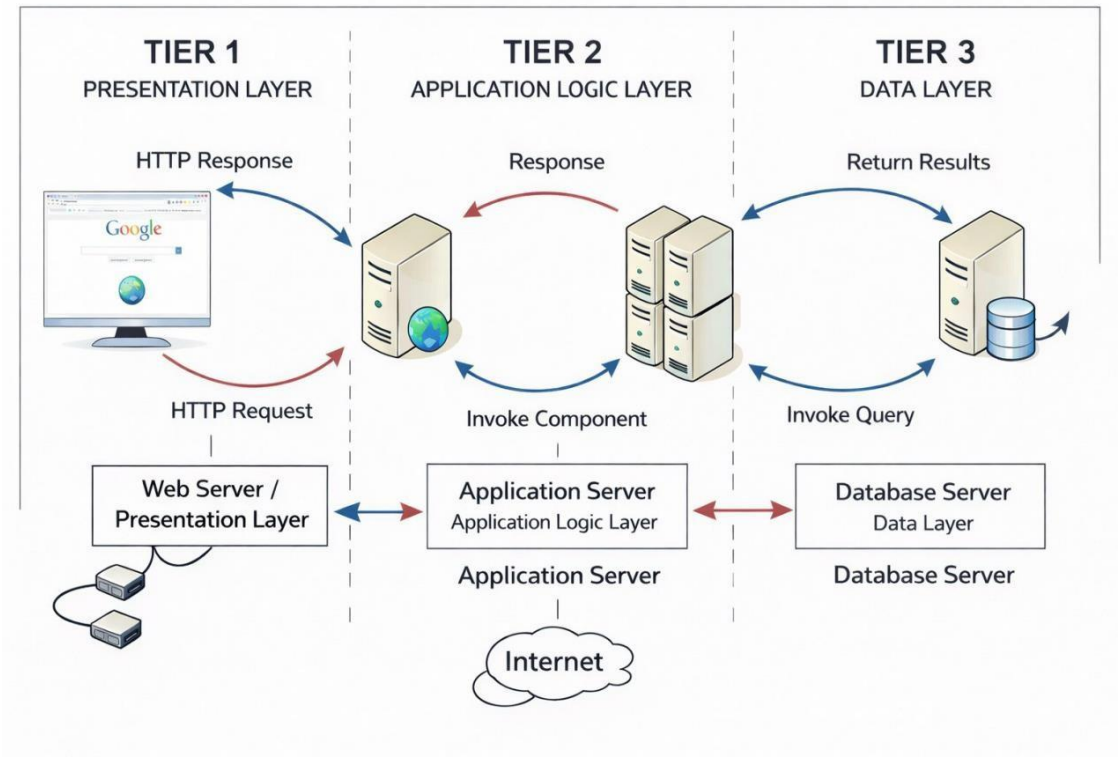

Figure 4.2: System Architecture Model for the Proposed System

4.3.1 System Decomposition

System decomposition is the process of breaking down the complex Web-Based Computer Laboratory Management System (CLMS) into smaller, more manageable units called subsystems or modules. Each subsystem is responsible for a specific set of functionalities, which helps in reducing the overall complexity of the design and ensures that different parts of the system can be developed and maintained independently. Based on the requirements and analysis models discussed in previous chapters, the CLMS is decomposed into several core subsystems, including User Management, Booking and Scheduling, Inventory and Asset Tracking, Maintenance and Fault Reporting, and the Reporting and Analytics module. The User Management Subsystem handles all aspects of user identity, including registration and role-based access control for students, teachers, and admins. The Booking and Scheduling Subsystem manages the reservation of laboratory rooms and specific workstations, utilizing automated logic to prevent double-bookings. To monitor physical resources, the Inventory and Asset Tracking Subsystem maintains a digital record of all hardware assets across the Maraki, Atse Tewodros, Fasil, and GC campuses. Furthermore, the Maintenance and Fault Reporting Subsystem digitizes the repair process by allowing users to submit tickets for broken equipment, while the Reporting and Analytics Subsystem aggregates data from all modules to generate real-time administrative insights. By decomposing the system into these functional units, we ensure high cohesion within each module and low coupling between them, making the system easier to implement, test, and scale according to the University of Gondar's needs.

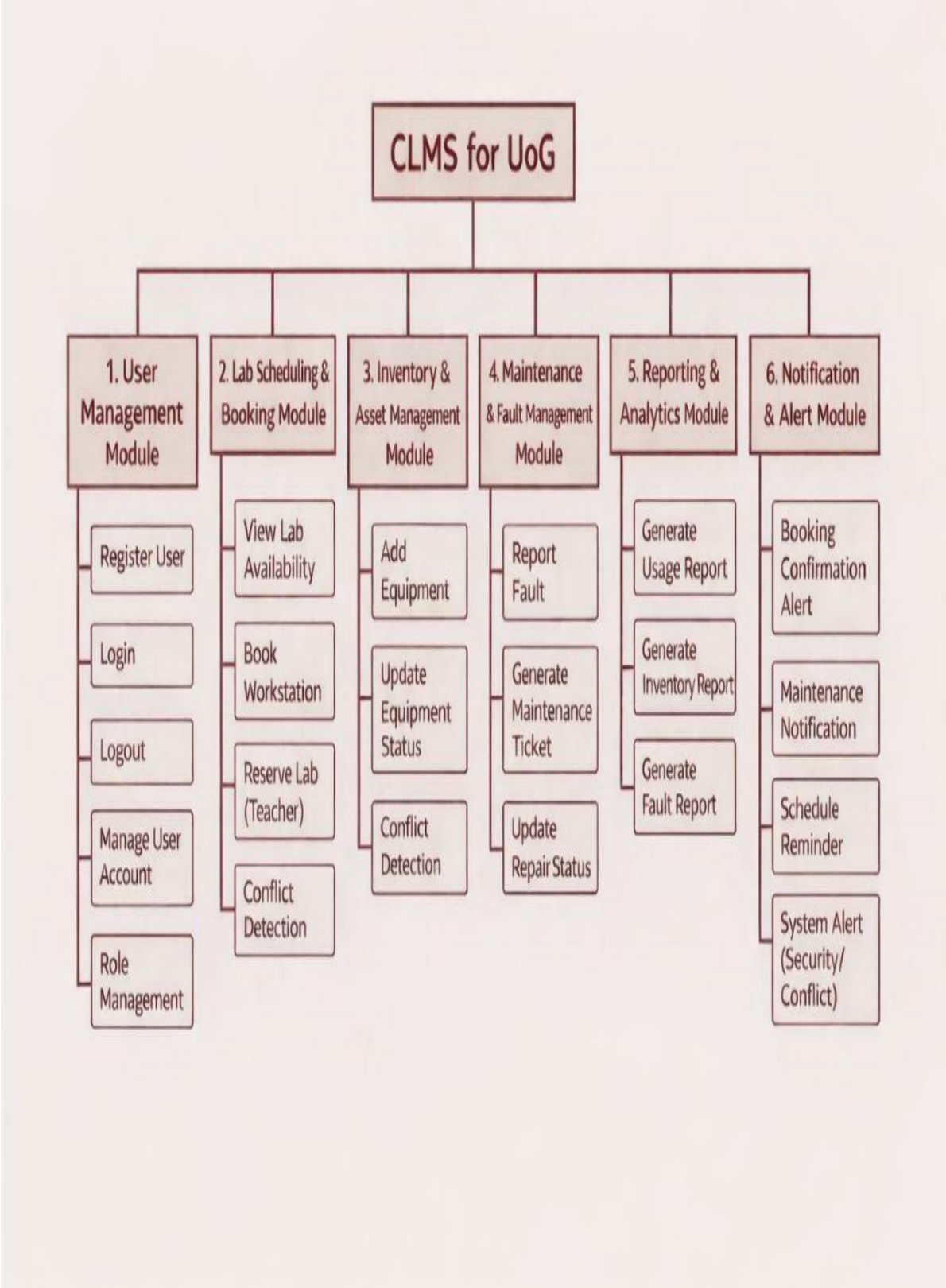

Figure 4.3**:** Decomposition Diagram

4.3.2 Component Diagram

It may be built from components in a component-based architecture. A component diagram shows how objects (classes) in our system are grouped together to form components. The components interact with each other either in giving service to other components or requesting service from another component.

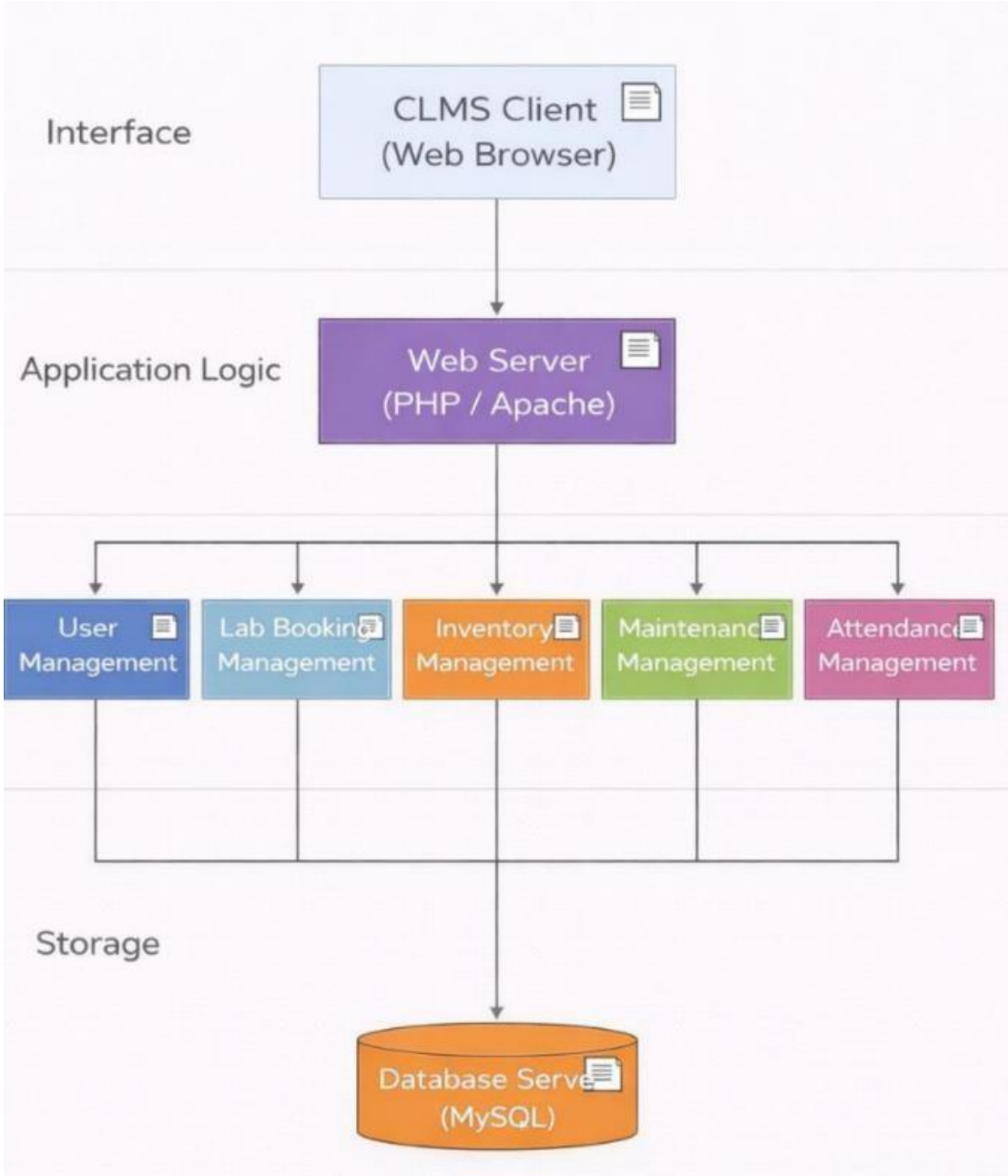

Figure 4.4**:** Component Diagram

4.3.3 Hardware/Software Mapping

Hardware/software mapping is the process of assigning the decomposed software subsystems and components to the physical hardware nodes and processors of the system. In the context of the Computer Laboratory Management System (CLMS), this mapping defines how the application layers, Presentation, Logic, and Data interact with the physical infrastructure of the university, including the central university server and the client devices used by students and staff. This mapping is essential for addressing performance goals, such as reducing response time, and reliability goals, by identifying potential points of failure and ensuring data persistence.

The software components, including the web-based user interface and the core application modules (Booking, Maintenance, and Inventory), are mapped to a central University Web Server. The relational database management system is mapped to a dedicated Database Server to ensure secure and optimized data handling. On the client side, the presentation layer is mapped to end-user devices such as desktop computers in the labs, laptops, and mobile devices, which access the system via the university's Local Area Network (LAN). By mapping the system in this manner, we can identify and manage potential concurrency, such as multiple students accessing the booking subsystem simultaneously from different campuses like Maraki or Fasil, ensuring that the hardware resources are utilized efficiently to meet the operational requirements of the CLMS.

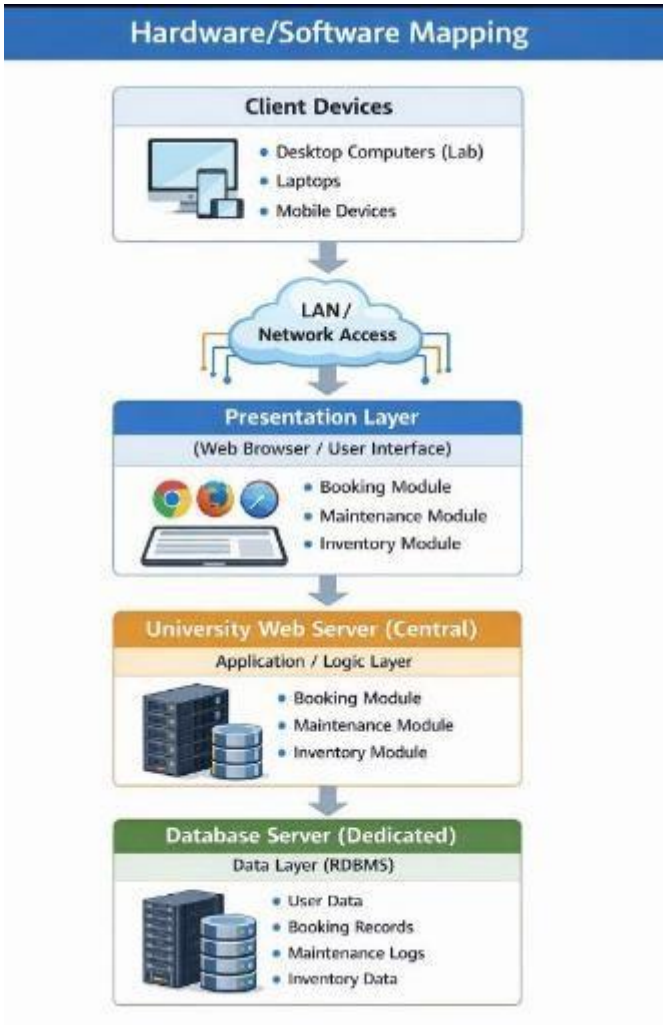

Figure 4.5: Hardware/Software Mapping

4.3.4 Persistent Data Management

Persistent data management is used to represent the design and organization of the system database. Database design is the process of creating a detailed data model that defines how data is stored, accessed, and maintained within the Computer Laboratory Management System (CLMS). The CLMS uses database tables to store important information that flows throughout the system.

The database contains tables for users, laboratories, workstations, bookings, maintenance records, and reports. These tables are structured using primary and foreign keys to establish relationships among different entities. This design ensures data consistency, minimizes redundancy, and supports efficient data retrieval. Persistent data management enables the system to store data permanently so that information remains available even after the system is shut down or restarted.

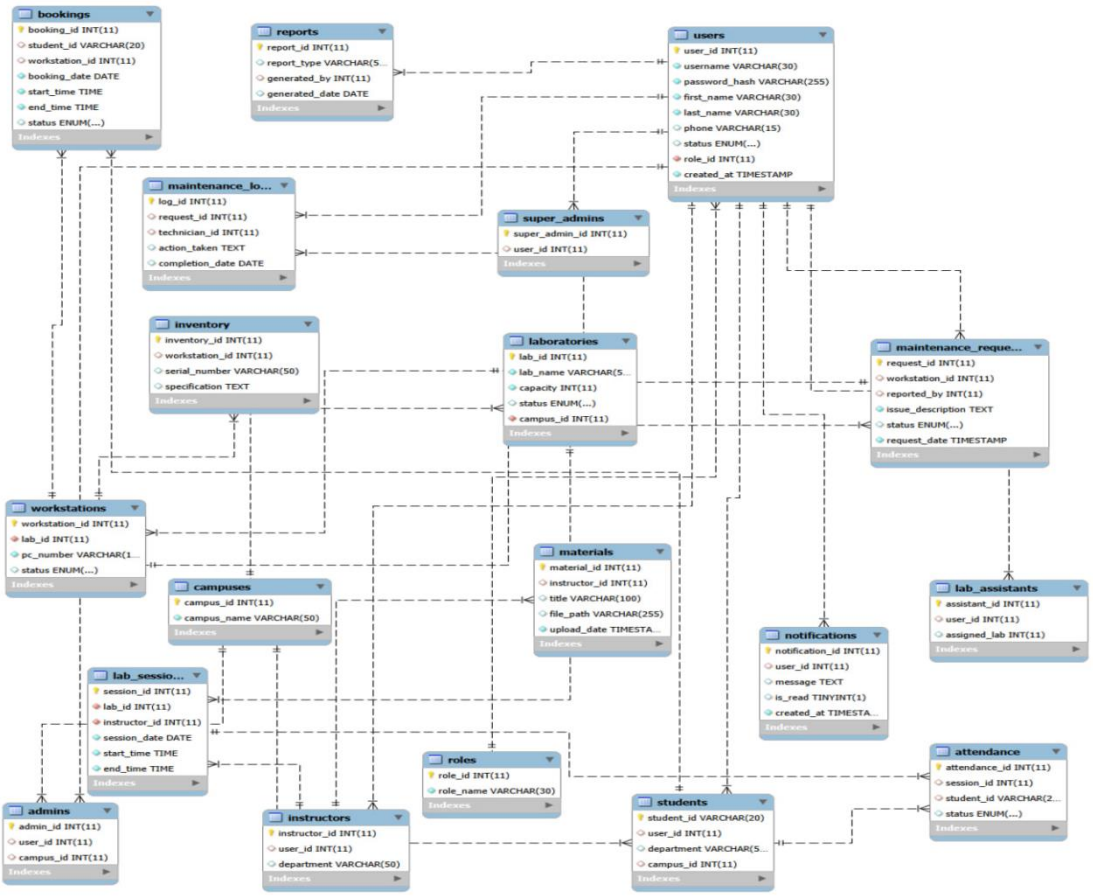

Figure 4.6: Database Schema

4.3.5 Access Control and Security

Access control and security define the mechanisms used to protect the Computer Laboratory Management System (CLMS) from unauthorized access while ensuring that legitimate users can perform their tasks based on their assigned privileges. Given that the system handles sensitive university data, including student records, staff credentials, and expensive hardware inventory across the Maraki, Atse Tewodros, Fasil, and GC campuses, a robust security framework is essential. The system employs Role-Based Access Control (RBAC), which maps specific permissions to user roles rather than individuals. For instance, Students are granted read and "create" permissions for workstation bookings, while Teachers have additional view permissions for lab schedules. Lab Assistants are granted update permissions for maintenance tickets and

inventory status, and only the System Administrator has full control over user management and system configurations.

To ensure data integrity and confidentiality, the CLMS implements several security layers, including secure authentication, data encryption, and audit trails. All users must undergo a secure login process where credentials are encrypted before being transmitted to the database. Furthermore, session management is used to prevent unauthorized session hijacking, and input validation is enforced on all web forms to protect against common security threats such as SQL injection and Cross-Site Scripting (XSS). An audit log is also maintained to track significant actions, such as who modified an inventory record or who approved a lab reservation, ensuring accountability. By integrating these security measures into the design model, the CLMS satisfies the non-functional requirements of manageability and security defined in the SRS.

Use Case	Student	Teacher	Lab Technician	Admin	Super Admin
Login	✓	✓	✓	✓	✓
Registration	✓	✓	✕	✕	✕
View Lab Availability	✓	✓	✓	✓	✓
Book Workstation	✓	✕	✕	✕	✕
Reserve Lab	✕	✓	✕	✓	✓
Download Material	✓	✕	✕	✕	✕
Upload Material	✕	✓	✕	✕	✕
Report Fault	✓	✓	✓	✓	✓
Monitor Attendance	✕	✓	✕	✕	✕
Check Equipment Status	✕	✕	✓	✓	✓
Manage Inventory	✕	✕	✓	✓	✓
Conflict Detection	✕	✕	✕	✓	✓
Manage User Account	✕	✕	✕	✓	✓

Generate Report	✕	✕	✓	✓	✓
Logout	✓	✓	✓	✓	✓

Table 4.1: Access Control and Security

4.3.6 Package Diagram

The Computer Laboratory Management System (CLMS) follows a three-tier architectural style. The subsystems of the system are identified in the previous sections, and their dependencies are also described. In this subsection, packages are drawn for each subsystem, and related subsystems are grouped into three main packages.

➢ Interface Package Layer:

This package represents the client tier and is responsible for providing the user interface of the system. It allows users such as students, instructors, lab assistants, and administrators to interact with the CLMS. Through this package, users can initiate system use cases such as viewing laboratory schedules, booking workstations, reporting faults, and generating reports.

➢ Application Logic Package Layer:

This package represents the middle tier of the system and contains sub-packages such as the server and data management packages. The server package is responsible for handling requests received from the interface package, processing business logic, and sending responses back to the client. The data management package is used by the server package to access, manipulate, and manage data stored in the storage package.

➢ Storage Package Layer:

This package represents the data tier of the system and consists of the database. It is responsible for storing persistent objects and system information such as user data, laboratory details, workstation information, booking records, maintenance data, and reports. The storage package ensures data persistence and integrity within the CLMS.

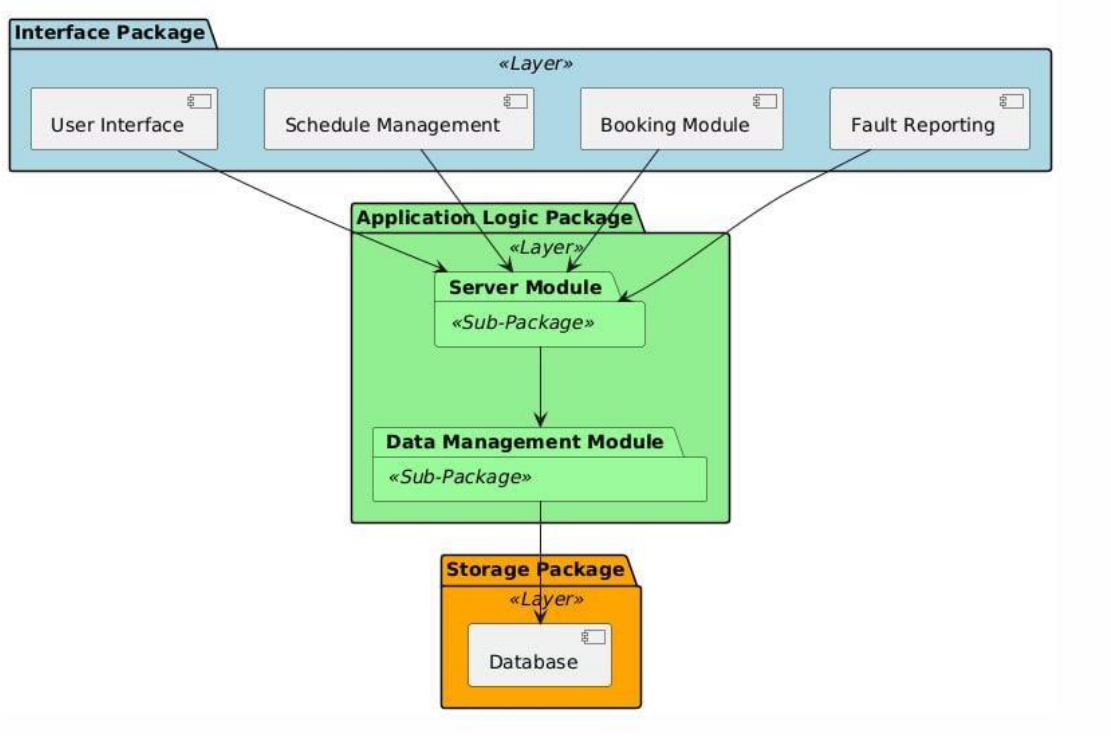

Figure 4.7**:** Package Diagram

4.3.7 Deployment Diagram

The deployment diagram illustrates the physical architecture of the Web-Based Computer Laboratory Management System (CLMS), showing the distribution of software components across hardware nodes and the communication paths between them. The system follows a three-tier client-server architecture consisting of three main nodes: Client Machine, Web Server Machine, and Database Server Machine.

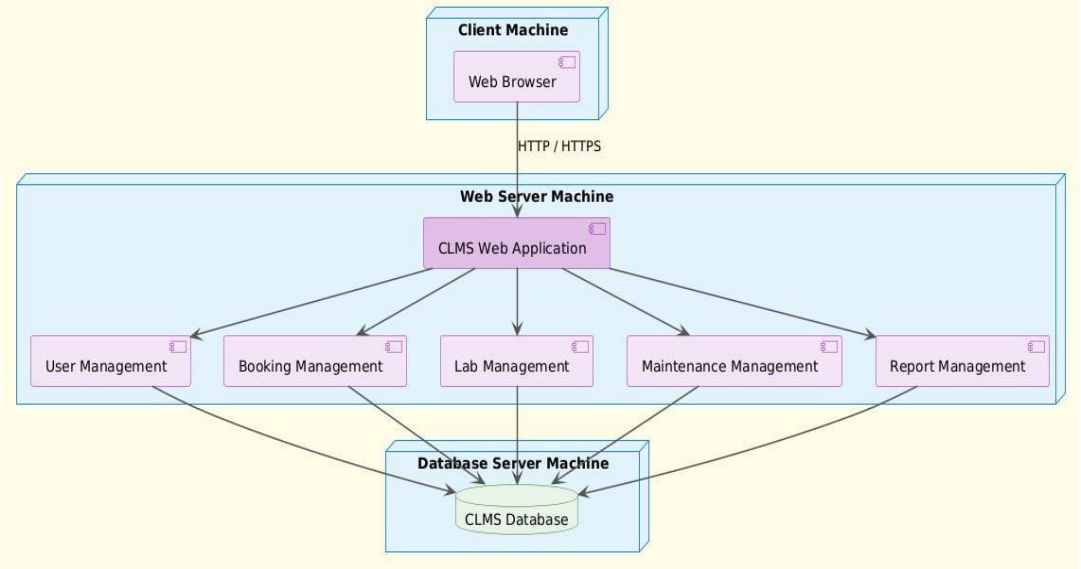

Figure 4.8: Deployment Diagram

CHAPTER FIVE: CONCLUSION AND RECOMMENDATION

CONCLUSION

The Web-Based Computer Laboratory Management System (CLMS) developed for the University of Gondar successfully addresses the critical challenges of the existing manual, paper-based system across four campuses (Maraki, Atse Tewodros, Atse Fasil, and GC). The current system suffers from scheduling conflicts, equipment mismanagement, data loss, and inefficient resource utilization.

The proposed system introduces six core modules: User Management, Lab Scheduling & Booking, Inventory & Asset Management, Maintenance & Fault Management, Reporting & Analytics, and Notification & Alerts. Built using a three-tier architecture with PHP and MySQL, the system provides role-based access control for students, instructors, lab assistants, administrators, and super administrators.

Key features include real-time workstation booking, conflict detection, equipment tracking, digital fault reporting, automated attendance monitoring, and comprehensive reporting. The system is technically feasible, operationally practical, and economically viable using open-source technologies.

Ultimately, the CLMS will optimize the use of over 1,670 computers across 33 laboratories, reduce equipment downtime, prevent scheduling conflicts, and support the university's vision of becoming a top African research university by 2030.

RECOMMENDATION

According to the scope of the project, the team developed a Web-Based Computer Laboratory Management System for the University of Gondar. During the development of this system, the team faced several challenges such as limited project development resources, especially time constraints, and some technical difficulties during system implementation. However, through strong cooperation among the group members and continuous support from the project advisor, the team was able to successfully complete the system and achieve the intended objectives.

Even though the system has been successfully developed, there are still additional features that can be included in the future to further improve the efficiency and usability of the system once it is put into operation. With additional time and technological support, the system can be expanded and enhanced by integrating new technologies and functionalities.

Generally, the team recommends that the following improvements be considered in the future:

Developing a mobile application version: Create a mobile version of the system to allow users to access laboratory services using smartphones. Adding email or SMS notifications: Implement functionality to inform students and instructors about laboratory schedules, reservations, and announcements automatically.

Integrating with the University Student Information System (SIS): Link the system with the existing university database for better data sharing and streamlined user authentication. Implementing automatic backup and enhanced security: Establish automated data protection mechanisms to ensure long-term system reliability and data integrity.

REFERENCES

1. A. Dennis, B. H. Wixom, and D. Tegarden, Systems Analysis and Design: An Object-Oriented Approach with UML, 5th ed. John Wiley & Sons, 2015.

2. M. Fowler, UML Distilled: A Brief Guide to the Standard Object Modeling Language, 3rd ed. Addison-Wesley Professional, 2004.

3. R. S. Pressman, Software Engineering: A Practitioner's Approach, 8th ed. McGraw-Hill Education, 2014.

4. I. Sommerville, Software Engineering, 10th ed. Pearson Education, 2016.

5. L. Welling and L. Thomson, PHP and MySQL Web Development, 5th ed. Addison-Wesley Professional, 2016.

Appendix: Group Discussion Methods Requirements Validation

Objective**:**

To validate the current laboratory challenges and gather functional requirements through a direct consultative session with University of Gondar (UoG) technical staff and coordinators.

1. Participants of the Discussion:

Target Group**:** Lab Coordinators and Lab Assistants.
Location: College of Informatics (AtseTewodros Campuses).

Lab Coordinator Name: Mr.Amare

Lab Technicians Name: Mr.Lakachew and M.r: Geremew

2. Key Discussion Points:

Difficulties faced by students in finding available workstations across campuses.

The lack of a formal channel for reporting hardware and software faults.

The need for a centralized reservation system for instructors and departments.

3. Verification Sample:

Discussion Questions	Summary of Staff Responses
How is lab data currently managed?	We record everything in paper logbooks, which makes it very difficult to search for specific data or generate monthly reports.
How do students check for PC availability?	Students must visit the lab in person. If it is full, they often have to walk to another campus to try their luck there.
What is the biggest challenge with maintenance?	Faults are reported verbally. If the technician isn't there, the issue might be forgotten, and the PC remains out of service for weeks.
What features would help you most?	A digital dashboard to manage all labs at once and an automated way for students to report broken equipment.

Table 4.2: Verification Sample